UTM
UNIVERSITI TEKNOLOGI MALAYSIA

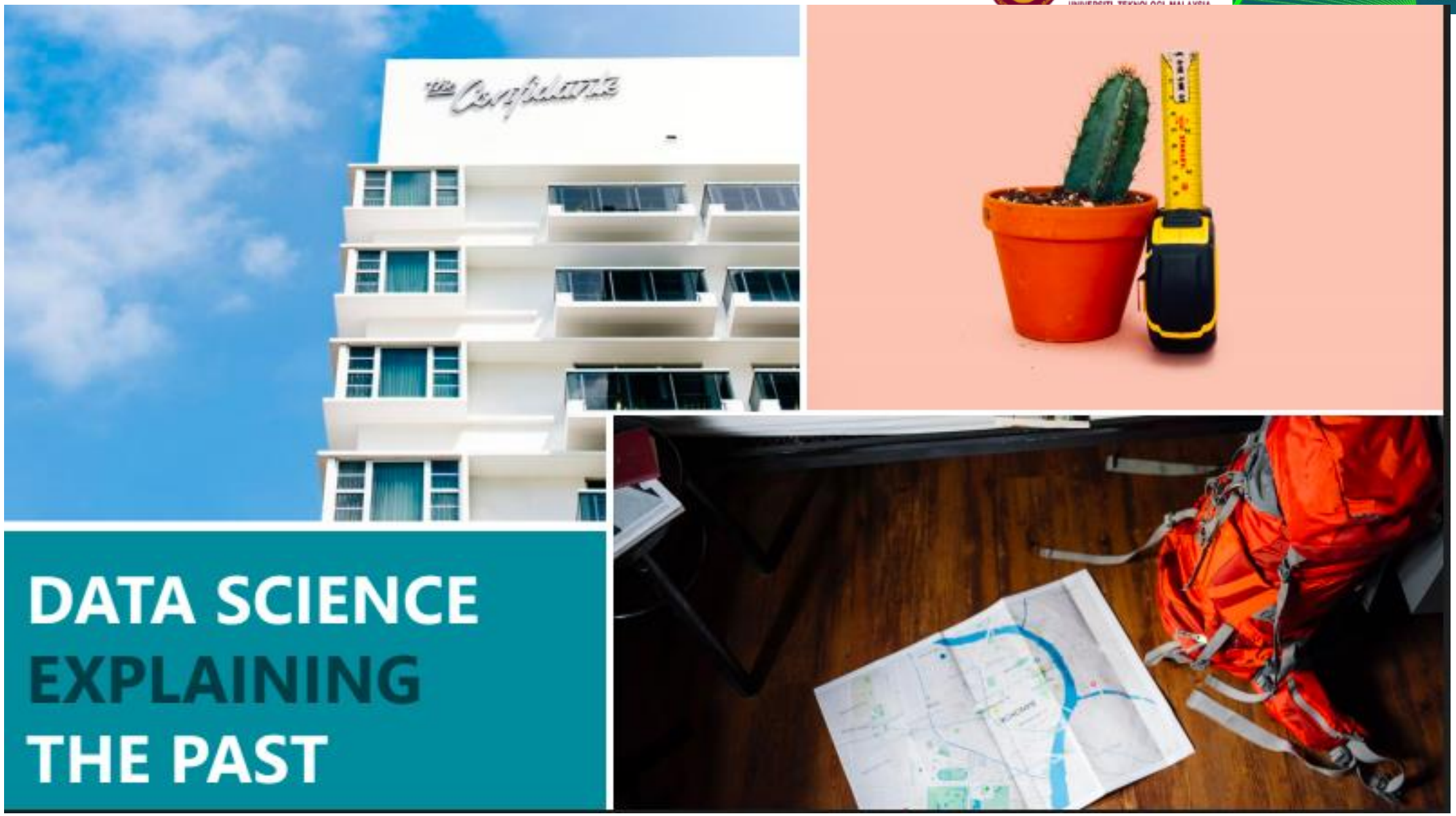
UTM BIG DATA CENTRE
TOGETHER WE MAKE MOTIONS...

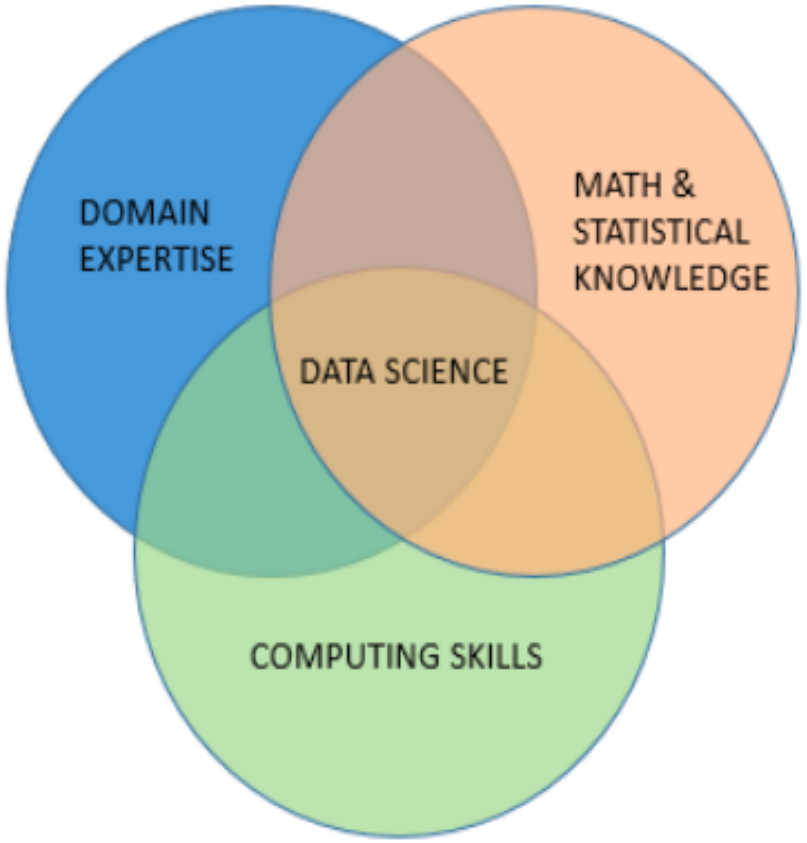
Introduction to Data Science: From Data to Insight


DATA

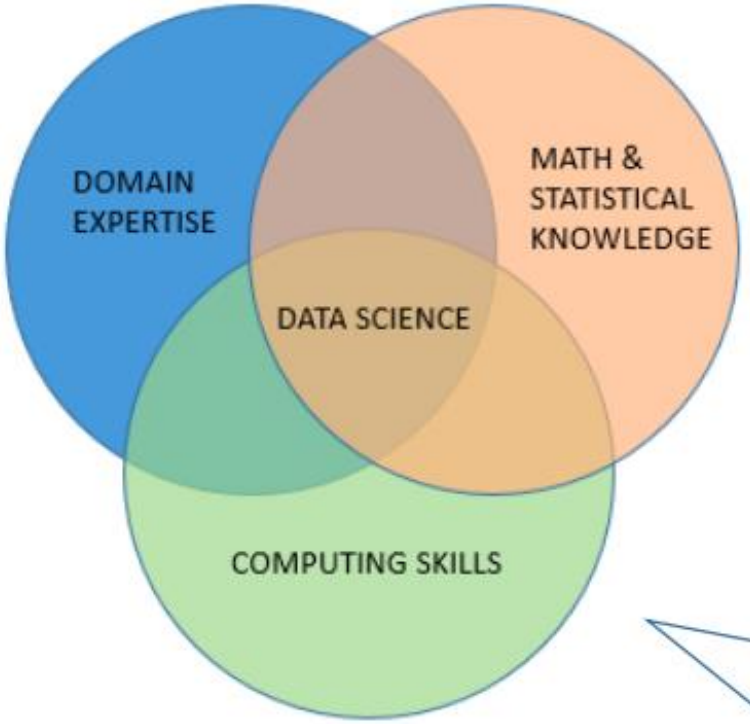
What is data?

- DATA is raw facts and figures.
- These have very little meaning until they are sorted or they are used to make calculations.
- The process of sorting or calculating data is called **DATA PROCESSING**.
- The result of data processing is **INFORMATION**.


DATA SCIENCE
EXPLAINING
THE PAST


DOMAIN EXPERTISE
MATH & STATISTICAL KNOWLEDGE
DATA SCIENCE
COMPUTING SKILLS


What does a data scientist do?
Raw Data
Processing
Dataset
Statistical Models / Analysis
Machine Learning Predictions
Data driven Products
Reports Visualization



Dealing with numbers, math formula and statistics, modelling

Dealing with business… and human!

Dealing with computing language, algorithms

Data science is a multi-disciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from structured and unstructured data. Wikipedia

WHAT IS DATA SCIENCE?

TURNING DATA INTO **INFORMATION**

ANALYZING DATA TO GET INSIGHTS

IDENTIFYING **TRENDS, PATTERNS,** AND **CORRELATIONS**

CONTEXTUALIZING, APPLYING AND **UNDERSTANDING** THEM

WHAT DOES A DATA SCIENTIST DO?




Identify organizational pinch points…
Maintain solid partnerships with your Finance and IT departments.
If you're not measuring, start tomorrow!!
Measuring is a journey-not a destination….

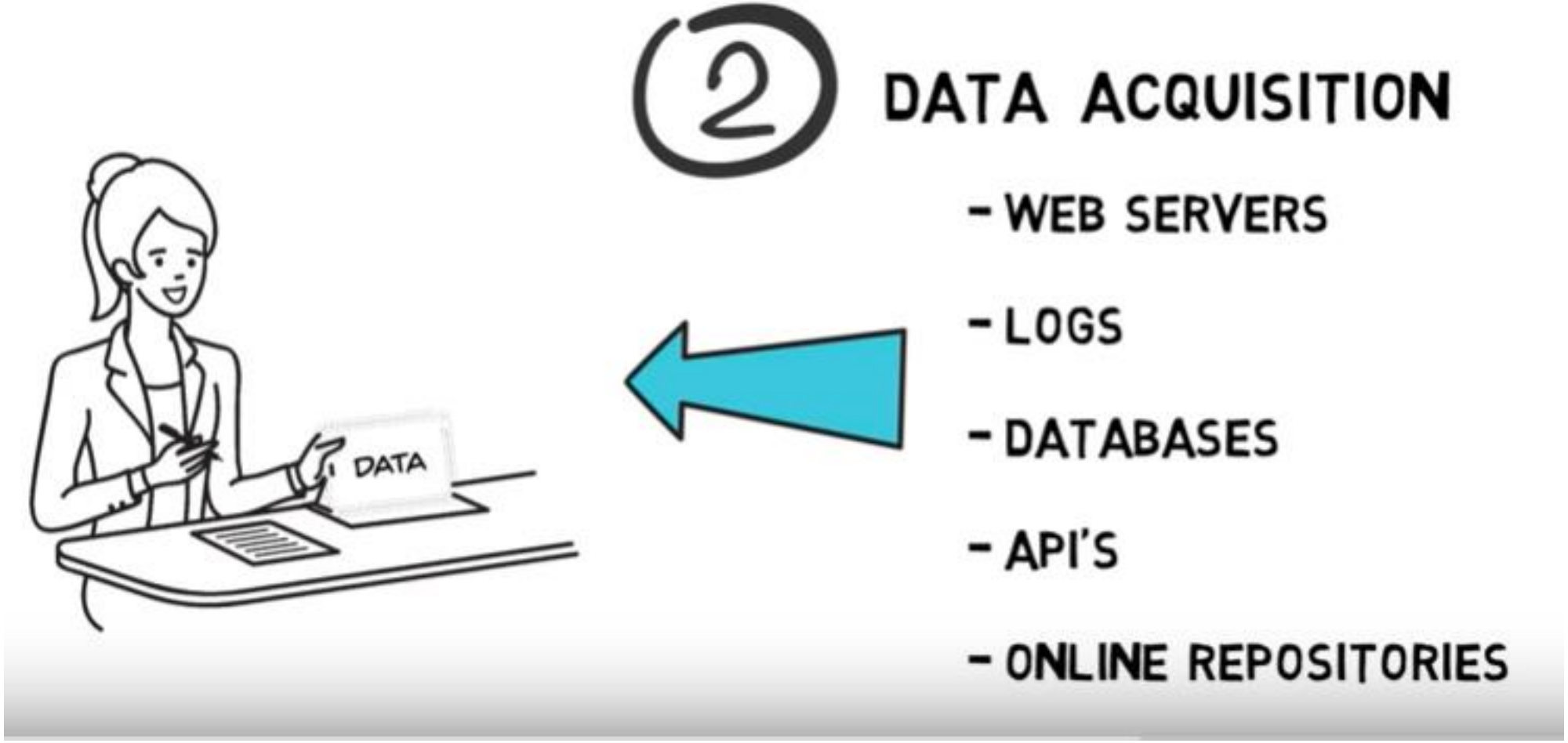
Translate data questions into something actionable

1.Who are the customers?
2.Why are they buying our product?
3.How do we predict if a customer is going to buy our product?
4.What is different from segments who are performing well and those that are performing below expectations?
5.How much money will we lose if we don't actively sell the product to these groups?



Need to understand why certain segments of customers have bought less than expected.

The end goal might be to determine whether to continue to invest in these segments, or de-prioritize


2
DATA ACQUISITION
- WEB SERVERS
- LOGS
- DATABASES
- API'S
- ONLINE REPOSITORIES
DATA

Collect the raw data needed for your problem

- ✓ Once you've defined the problem,
- ✓ you'll need data to give you the insights needed to turn the problem around with a solution.
- ✓ This part of the process involves thinking through what data you'll need and finding ways to get that data, whether it's
  - ➢ querying internal databases, or
  - ➢ (purchasing) external datasets

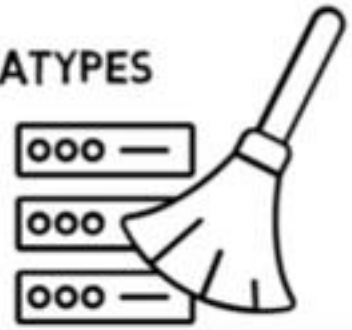


DATA PREPARATION

DATA CLEANING

INCONSISTENT DATATYPES

MISSPELLED ATTRIBUTES



MISSING AND DUPLICATE VALUES

TRANSFORMATION



talend

Informatica

(4) EXPLORATORY DATA ANALYSIS

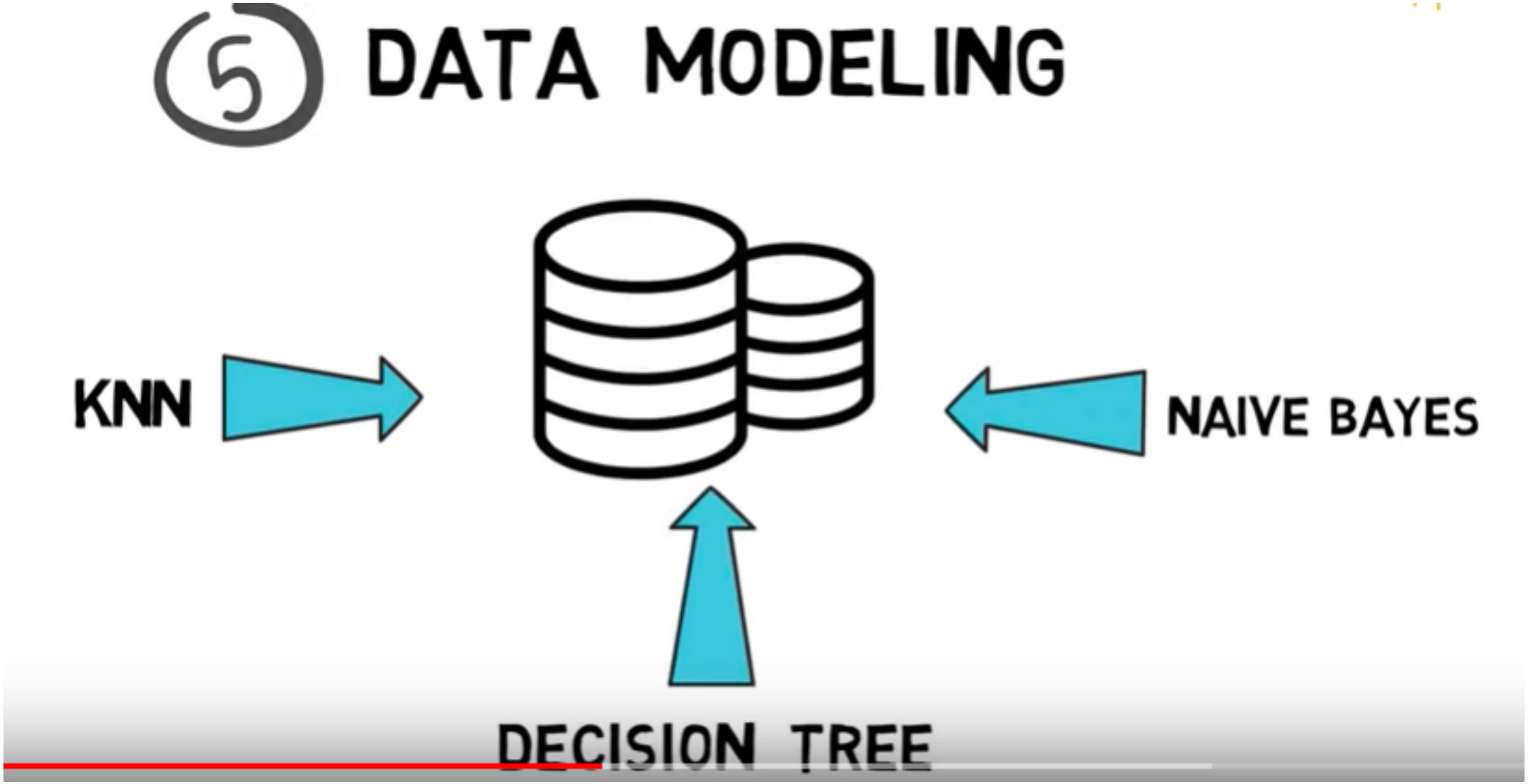


DEFINES AND REFINES

THE SELECTION OF FEATURE

VARIABLES THAT WILL BE USED

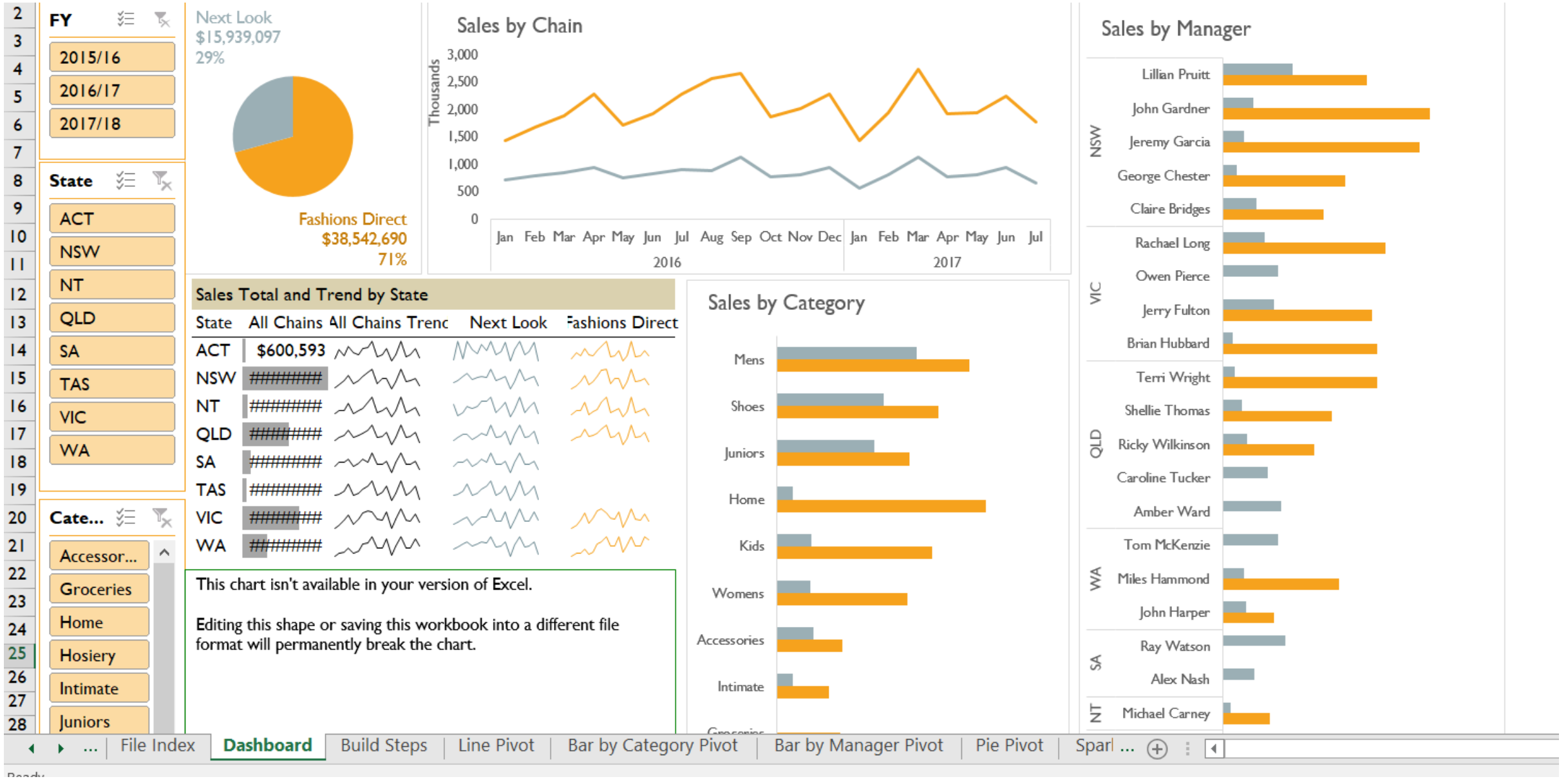
IN THE MODEL DEVELOPMENT

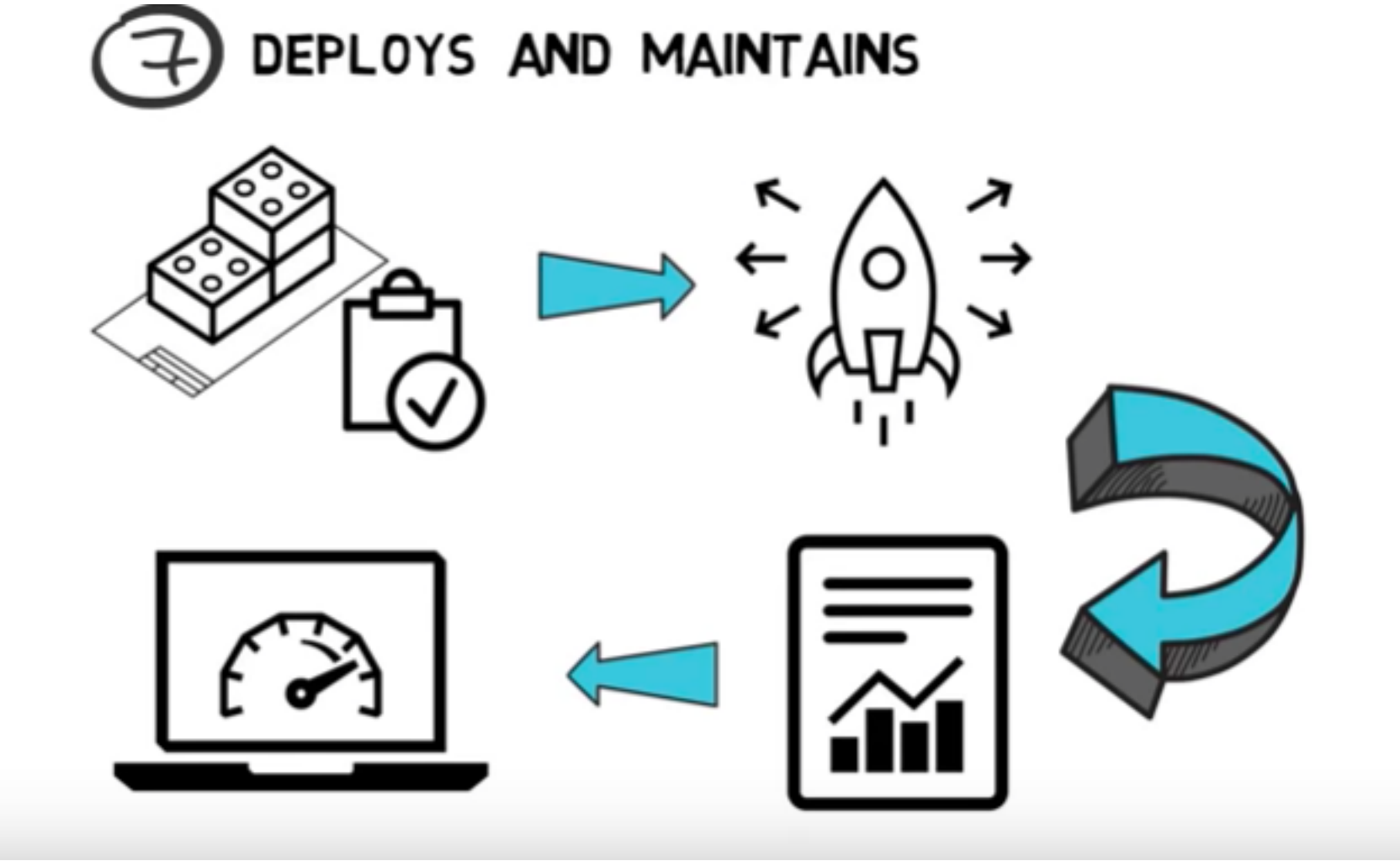

5
DATA MODELING
KNN
NAIVE BAYES
DECISION TREE

6 VISUALIZATION AND COMMUNICATION



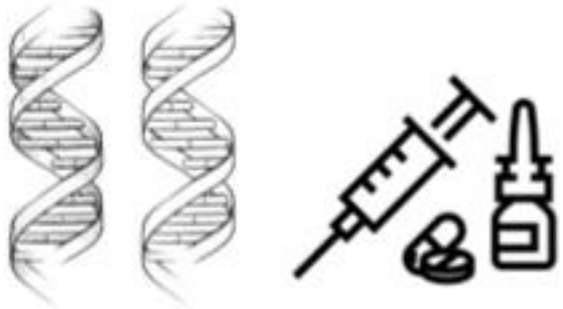
Craft a compelling story that ties your data with the knowledge




7
DEPLOYS AND MAINTAINS

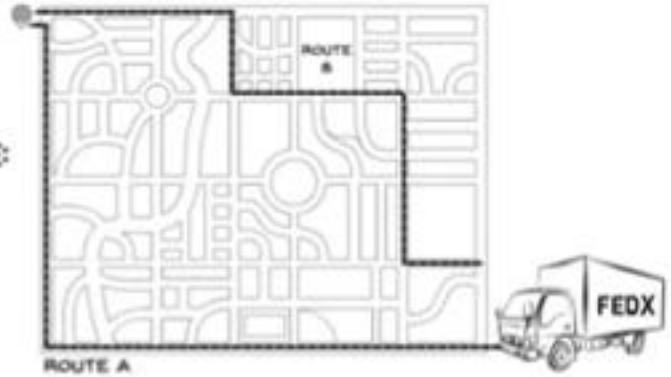
DS Solutions

GENOMIC DATA PROVIDES A DEEPER UNDERSTANDING OF GENETIC ISSUES



LOGISTICS COMPANIES LIKE DHL, FEDEX HAVE DISCOVERED THE BEST TIME AND ROUTES TO SHIP

BEST TIME: 12:00PM



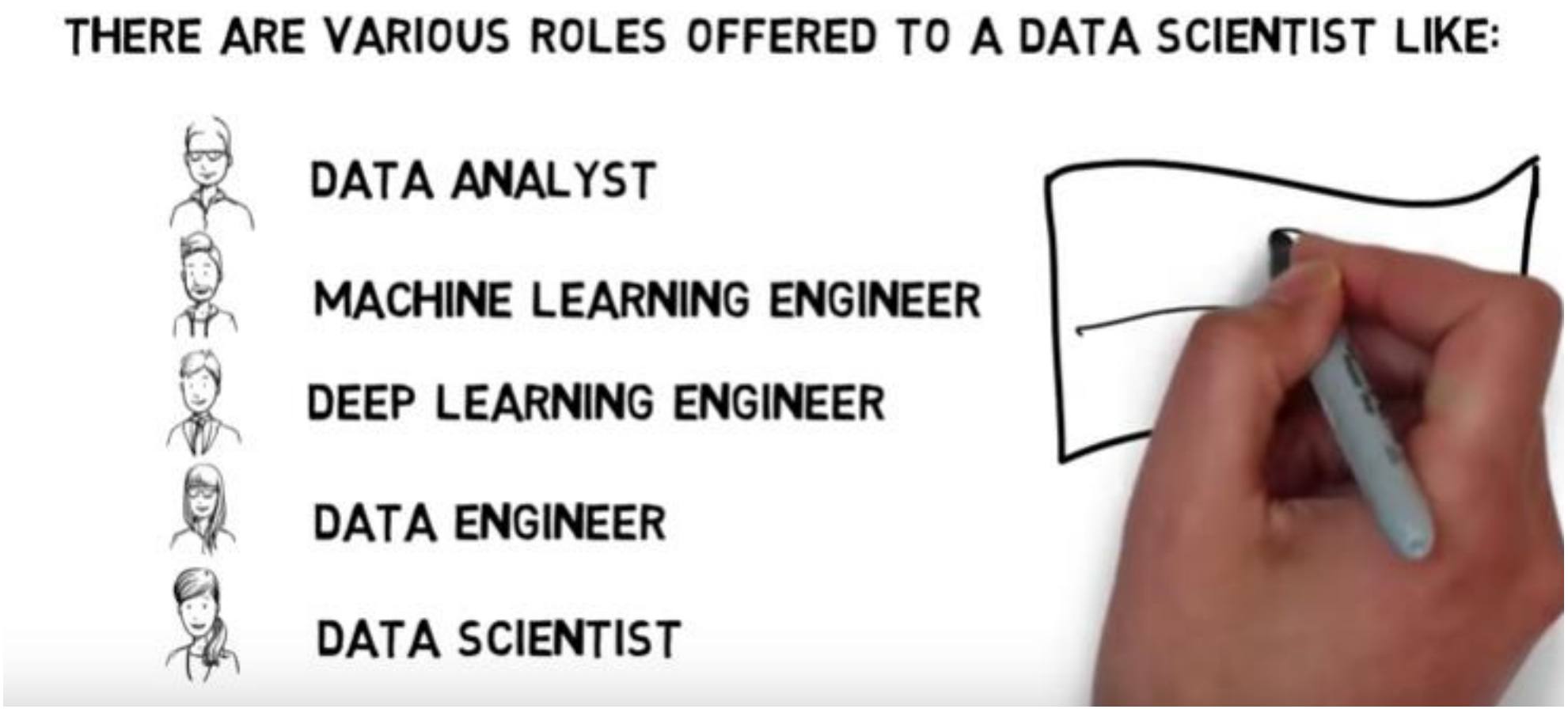
PREDICT EMPLOYEE ATTRITION AND UNDERSTAND THE VARIABLES THAT INFLUENCE EMPLOYEE TURNOVER

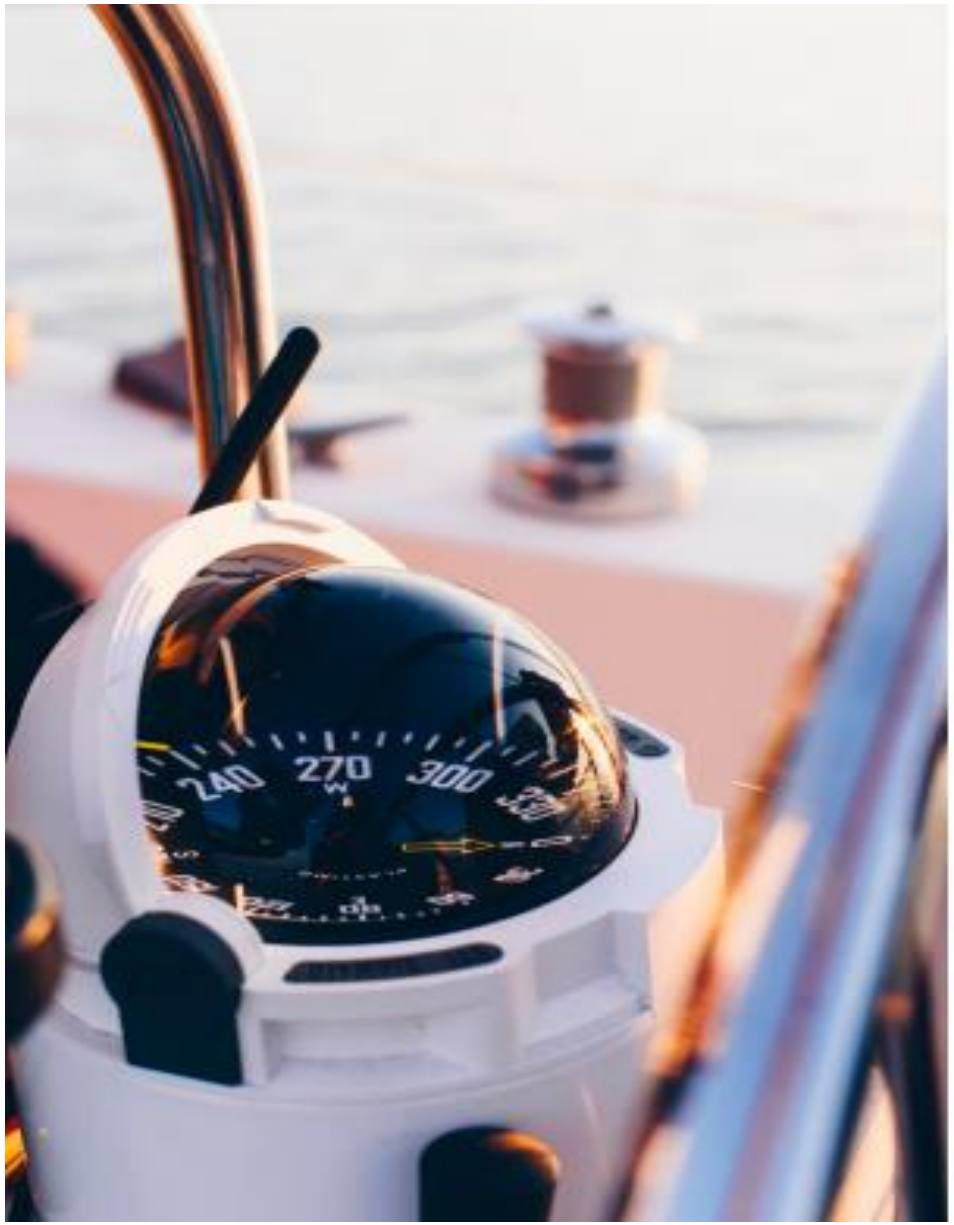


AIRLINE COMPANIES CAN NOW EASILY PREDICT FLIGHT DELAY AND NOTIFY THE PASSENGERS



H487	CANCELLED
PP87	DELAYED


THERE ARE VARIOUS ROLES OFFERED TO A DATA SCIENTIST LIKE:
DATA ANALYST
MACHINE LEARNING ENGINEER
DEEP LEARNING ENGINEER
DATA ENGINEER
DATA SCIENTIST



DATA SCIENCE PREDICTING *the future*

Predictive analytics in data science rest on the shoulders of explanatory data analysis.

In fact, everything is connected. Once the BI reports and dashboards have been prepared and insights – extracted from them – this information becomes the basis for predicting future values. And the accuracy of these predictions lies in the methods used.

Recall the distinction between traditional data and big data in data science. A similar distinction can be made regarding predictive analytics and their methods: traditional data science methods vs. Machine Learning. One deals primarily with traditional data, and the other – with big data.

365 DataScience

Data

Data Science

TRADITIONAL

BIG

BUSINESS INTELLIGENCE

TRADITIONAL METHODS

MACHINE LEARNING

WHEN *it is applied*

At the beginning of your analysis

After the data has been gathered & organized

After BI reports have been created and discussed

PAST — NOW — FUTURE

WHY *you need it*

data-driven decisions require well-organized and relevant raw data stored in a digital format

use data to create reports and dashboards to gain business insights

Predictive Analytics

assess potential future scenarios by using advanced statistical methods

utilize artificial intelligence to predict behavior in unprecedented ways

WHAT *techniques are involved*

DATA COLLECTION

PREPROCESSING

- class labeling *(categorical vs numerical)*
- data cleansing
- dealing with missing values

CASE SPECIFIC

- e.g. balancing & shuffling datasets





DATA COLLECTION

PREPROCESSING

- class labeling *(number, text, digital images, digital video data, digital audio data)*
- data cleansing
- dealing with missing values

CASE SPECIFIC

- text data mining, confidentiality - preserving data mining techniques



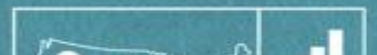
ANALYZE THE DATA

EXTRACT INFO AND PRESENT IT IN THE FORM OF:

- metrics
- KPIs
- reports
- dashboards







REGRESSION



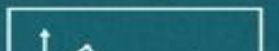
LOGISTIC REGRESSION

CLUSTERING



FACTOR ANALYSIS

TIME SERIES



SUPERVISED LEARNING

- SVMs
- NNs
- deep learning
- random forests
- bayesian networks

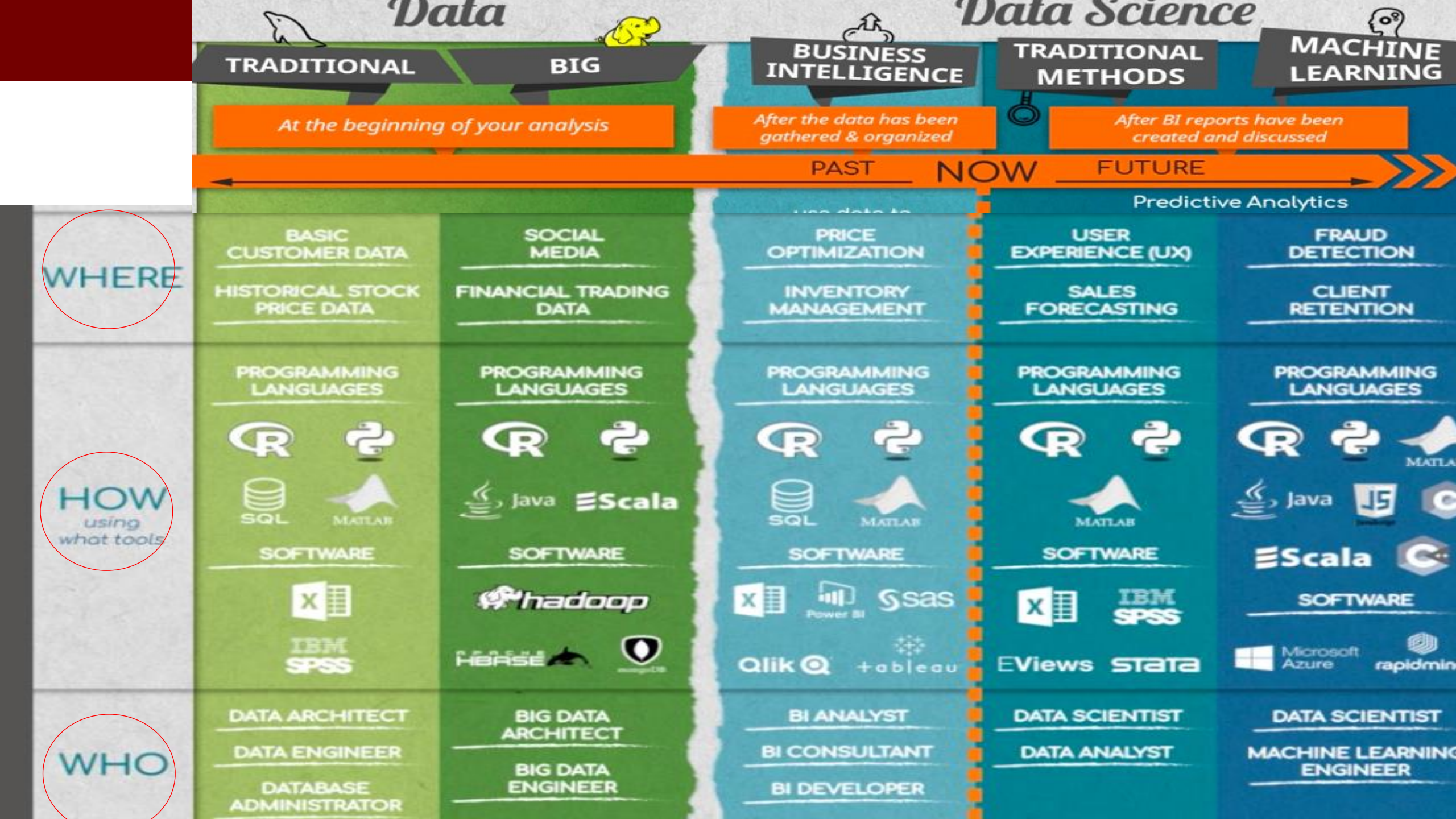
ML

UNSUPERVISED LEARNING

- k-means
- deep learning

REINFORCEMENT LEARNING

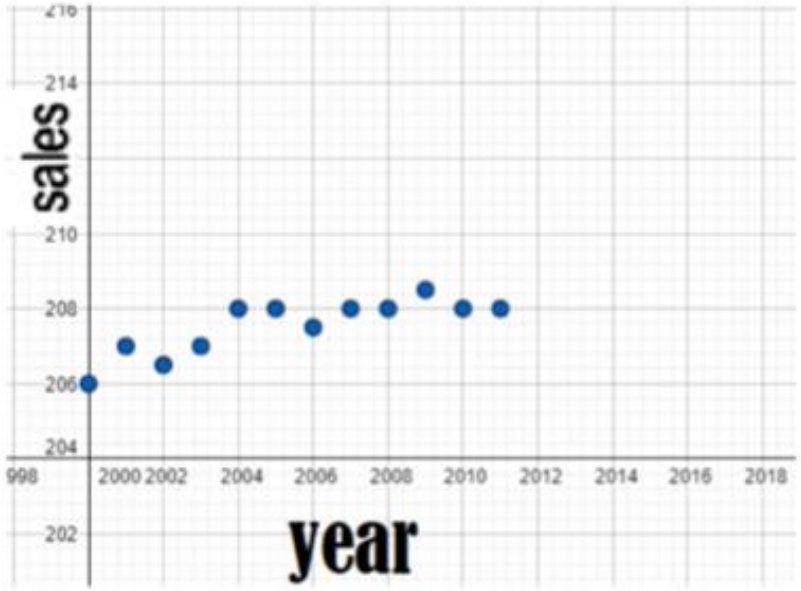
similiar to supervised learning, but instead of minimizing the loss,


Data
Data Science
TRADITIONAL
BIG
BUSINESS INTELLIGENCE
TRADITIONAL METHODS
MACHINE LEARNING
At the beginning of your analysis
After the data has been gathered & organized
After BI reports have been created and discussed
PAST
NOW
FUTURE
Predictive Analytics
WHERE
BASIC CUSTOMER DATA
HISTORICAL STOCK PRICE DATA
SOCIAL MEDIA
FINANCIAL TRADING DATA
PRICE OPTIMIZATION
INVENTORY MANAGEMENT
USER EXPERIENCE (UX)
SALES FORECASTING
FRAUD DETECTION
CLIENT RETENTION
HOW
using what tools
PROGRAMMING LANGUAGES
SQL
MATLAB
SOFTWARE
IBM SPSS
Java
Scala
hadoop
HBASE
Power BI
sas
Qlik
tableau
EViews
stata
Microsoft Azure
rapidmin
WHO
DATA ARCHITECT
DATA ENGINEER
DATABASE ADMINISTRATOR
BIG DATA ARCHITECT
BIG DATA ENGINEER
BI ANALYST
BI CONSULTANT
BI DEVELOPER
DATA SCIENTIST
DATA ANALYST
DATA SCIENTIST
MACHINE LEARNING ENGINEER

Skills Needed For Data Scientist and Data Analyst

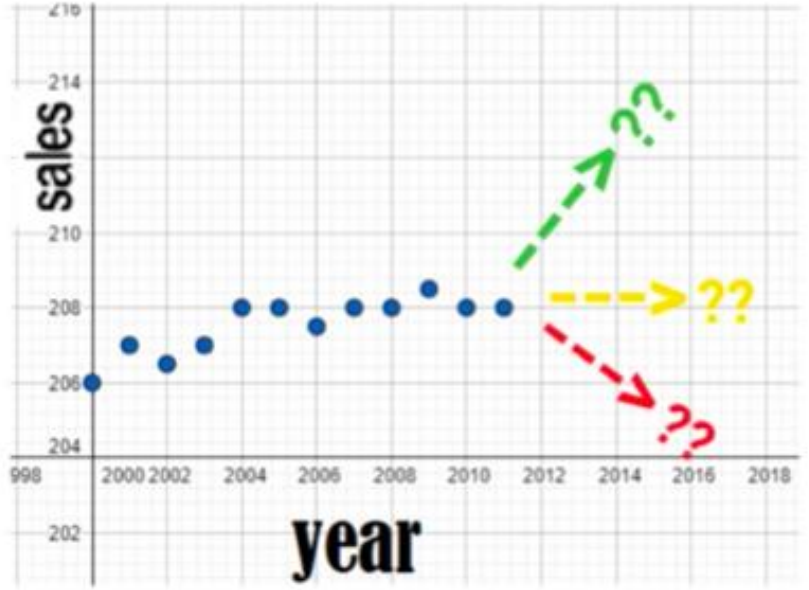
DATA ANALYST

DESCRIPTIVE ANALYTICS



DATA SCIENTIST

DESCRIPTIVE ANALYTICS
+ PREDICTIVE ANALYTICS



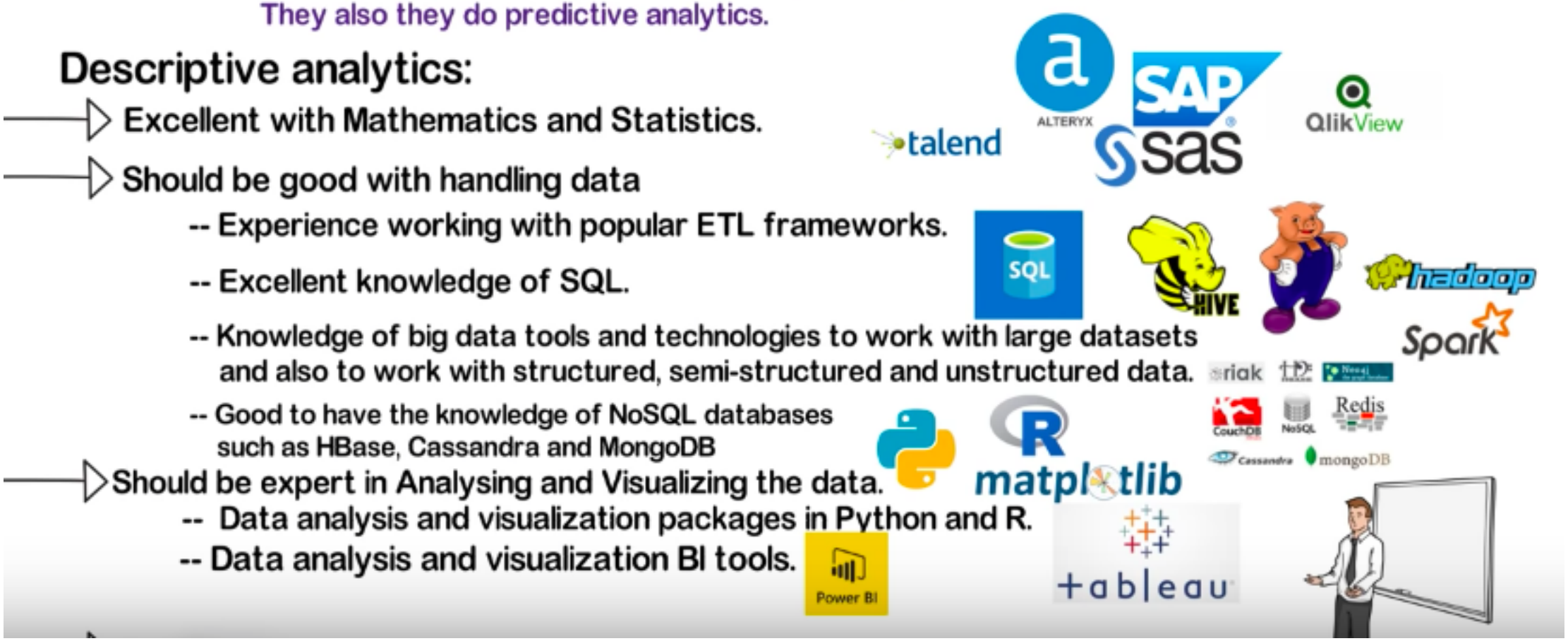
SKILLS NEEDED

DATA SCIENTIST:

Data scientist also does descriptive analytics just like data analysts.
They also they do predictive analytics.

Descriptive analytics:

- Excellent with Mathematics and Statistics.
- Should be good with handling data
  - -- Experience working with popular ETL frameworks.
  - -- Excellent knowledge of SQL.
  - -- Knowledge of big data tools and technologies to work with large datasets and also to work with structured, semi-structured and unstructured data.
  - -- Good to have the knowledge of NoSQL databases such as HBase, Cassandra and MongoDB
- Should be expert in Analysing and Visualizing the data.
  - -- Data analysis and visualization packages in Python and R.
  - -- Data analysis and visualization BI tools.



SKILLS NEEDED



DATA ANALYST :



→ Data Analysts should be good with Mathematics and Statistics.

→ They should be good with handling the data.

-- knowledge of ETL (Extract Transform and Load) operations on data and experience working with popular ETL tools.

-- comfortable in handling data from different sources and in different formats.

-- Excellent knowledge of SQL (or Structured Query Language).

Bonus to have:

-- knowledge of Big data tools and technologies

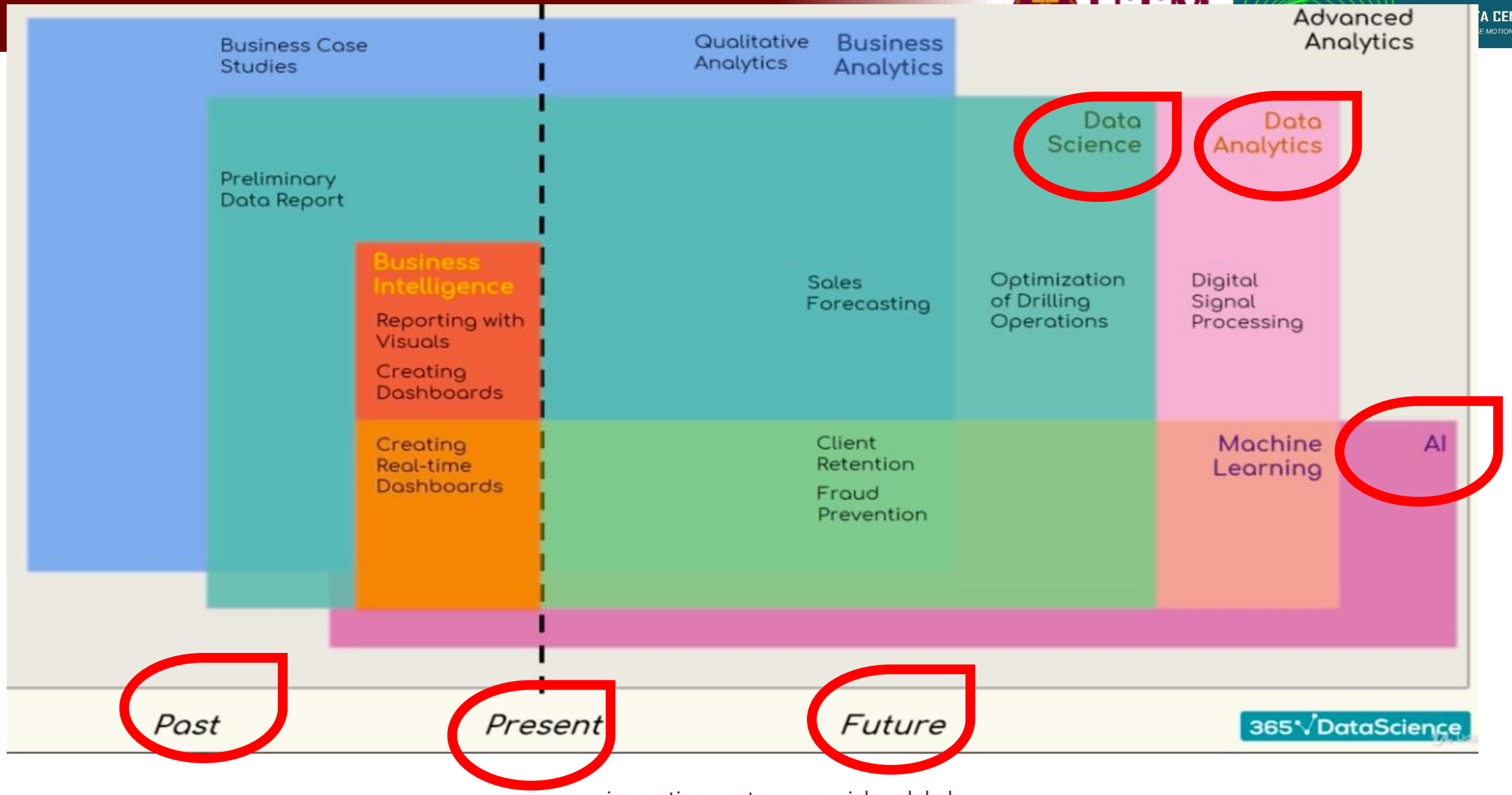
-- Experience with NoSQL databases such as HBase, Cassandra and MongoDB.

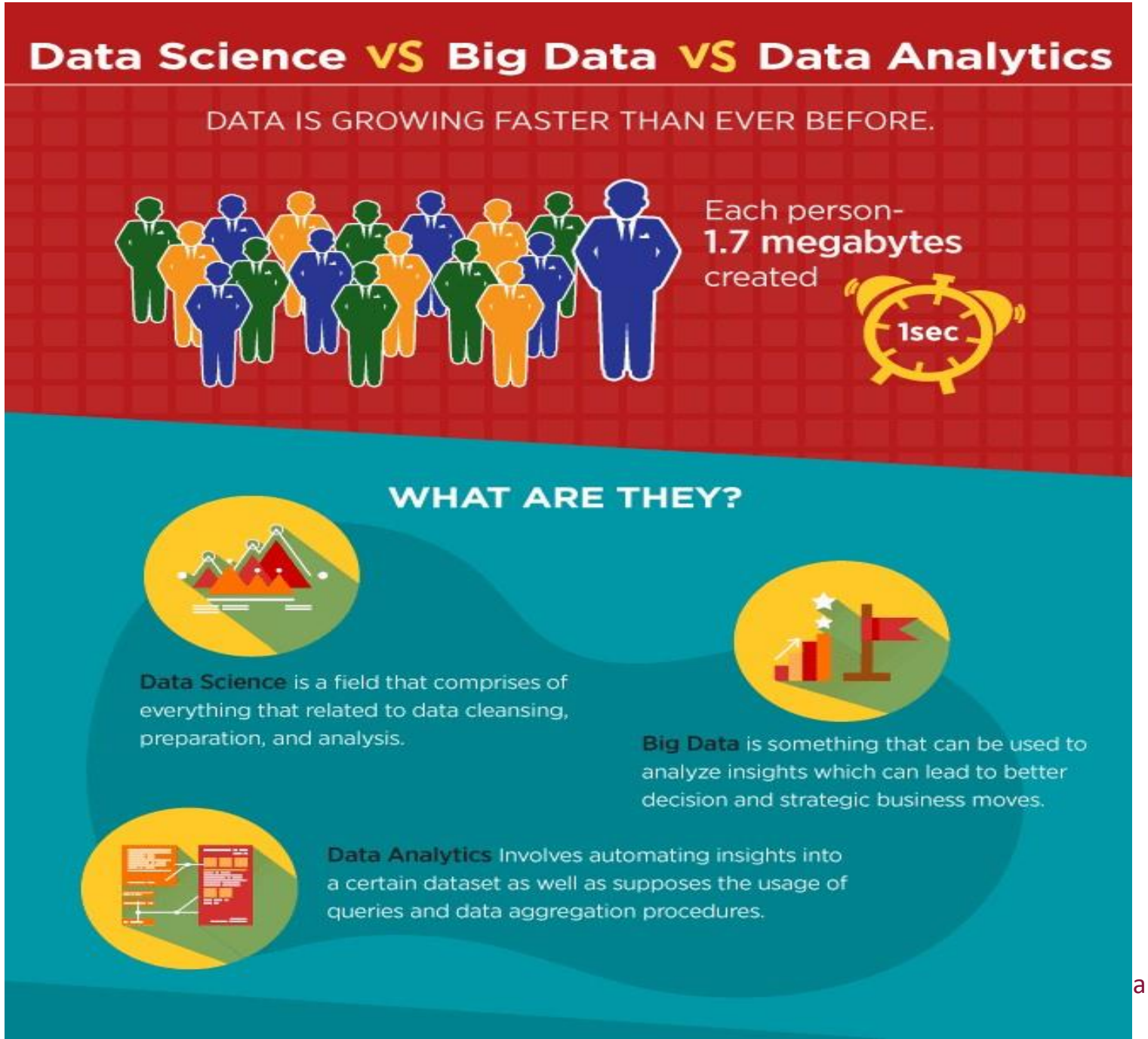
→ They should be expert in Analysing and Visualizing the data.

-- Experience working with popular data analysis and visualization packages in python and R such as numpy, scipy, pandas, matplotlib, ggplot and others.

-- Experience with popular data analysis and visualization tools such as Tableau, Microsoft Power BI, SAP BI, SAS BI, Oracle BI, QlikView or any other popular BI tool.

→ They should have good communication and storytelling skills.


Business Case Studies
Qualitative Analytics
Business Analytics
Advanced Analytics
Data Science
Data Analytics
Preliminary Data Report
Business Intelligence
Reporting with Visuals
Creating Dashboards
Sales Forecasting
Optimization of Drilling Operations
Digital Signal Processing
Creating Real-time Dashboards
Client Retention
Fraud Prevention
Machine Learning
AI
Past
Present
Future
365 DataScience



al https://www.simplilearn.com/data-science-vs-big-data-vs-data-analytics-article

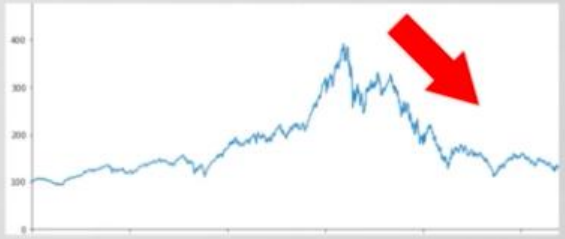
Analysis

Past



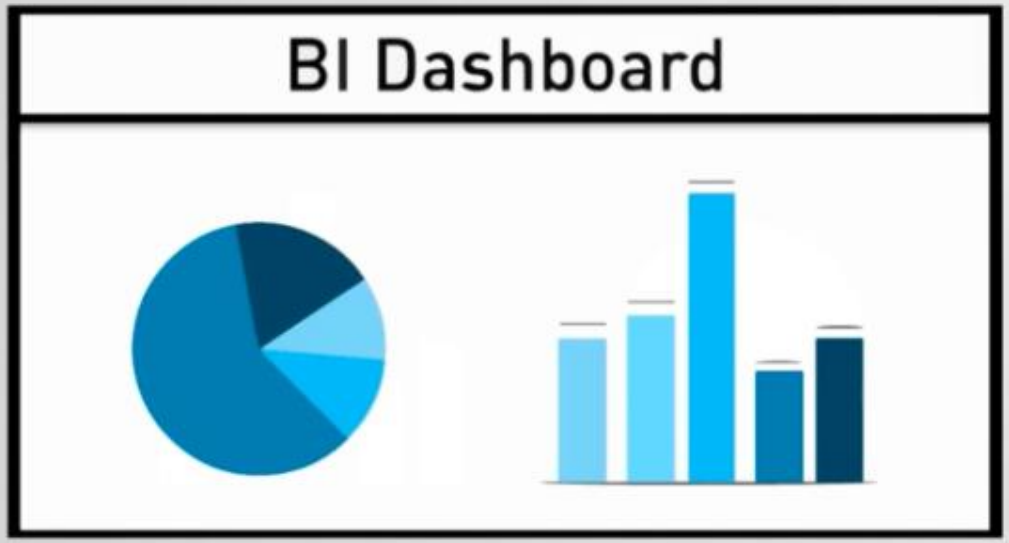
Explain

How? Why?



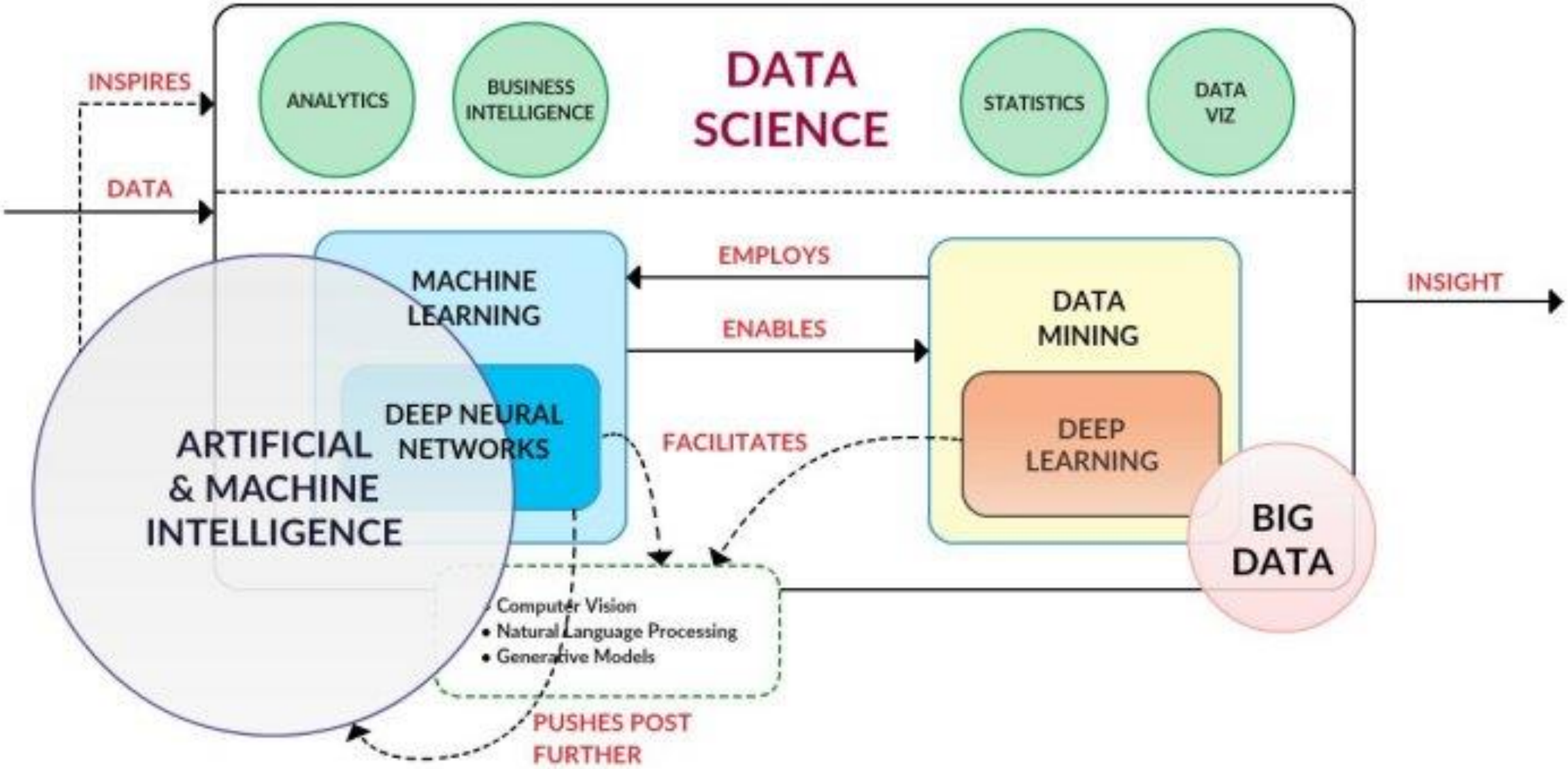
business intelligence (BI): *the process of analysing and reporting historical business data*

aims to explain past events using business data






Analytics
Future
Explore potential future events


INSPIRES
DATA
ANALYTICS
BUSINESS INTELLIGENCE
DATA SCIENCE
STATISTICS
DATA VIZ
MACHINE LEARNING
EMPLOYS
ENABLES
DATA MINING
INSIGHT
DEEP NEURAL NETWORKS
ARTIFICIAL & MACHINE INTELLIGENCE
FACILITATES
DEEP LEARNING
BIG DATA
• Computer Vision
• Natural Language Processing
• Generative Models
PUSHES POST FURTHER

DATA

information stored in a digital format, which can then be used as a base for performing analyses and decision making

Traditional data

- structured

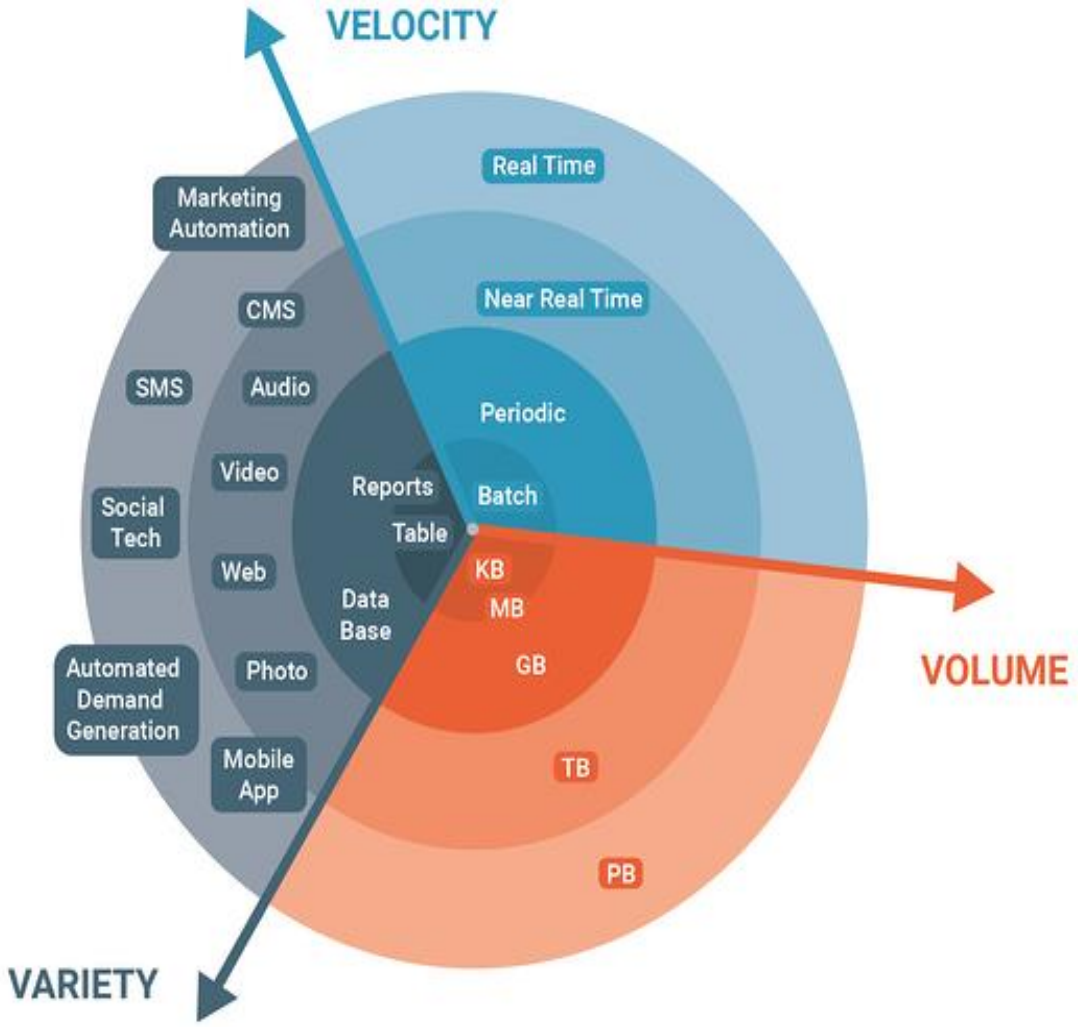
ID	Name	Age	.....
001	John	35	.....
002	Alan	22	.....
.....	.....	.....	.....

- 
can be managed from 1 computer

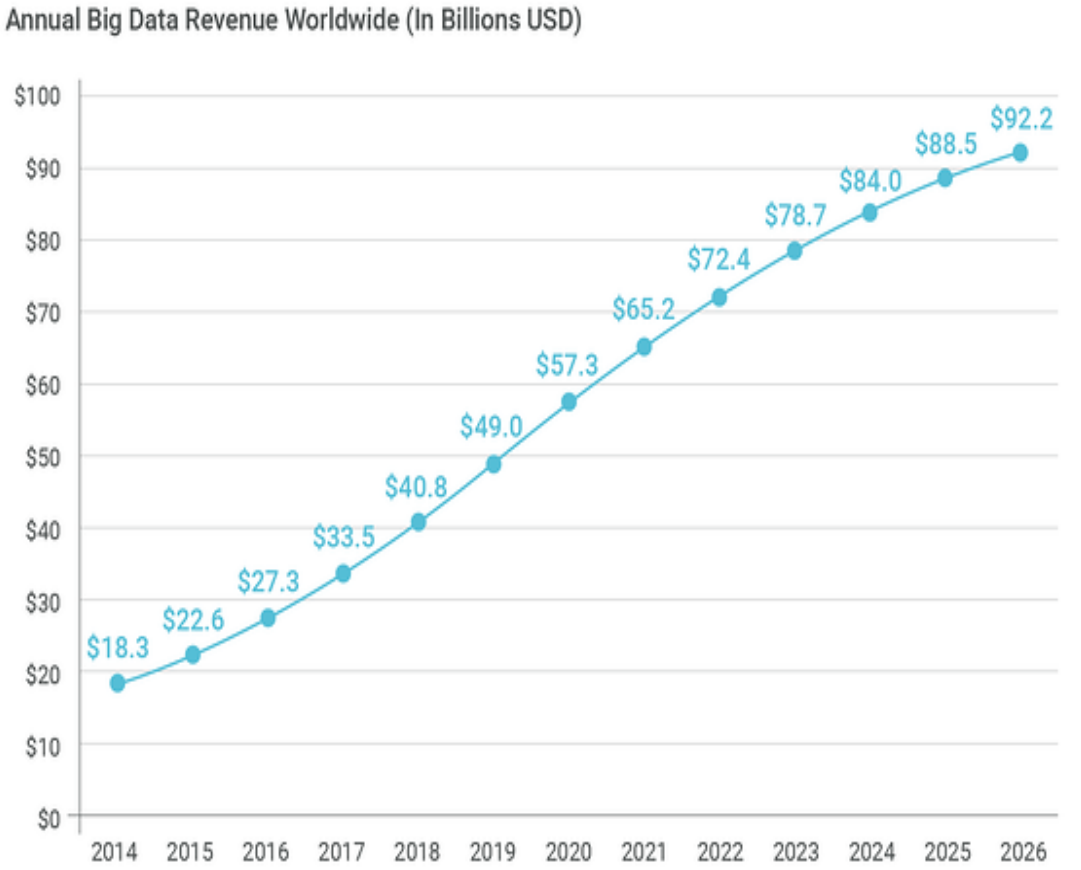


- Big data" describes innovative techniques and technologies to capture, store, distribute, manage and analyze datasets that **traditional data management methods are unable to handle** .
- Big data is larger, more complex data sets, especially from new data sources. These data sets are so voluminous that traditional data processing software just can't manage them. But these massive volumes of data can be used to address business problems you **wouldn't have been able to tackle before.**

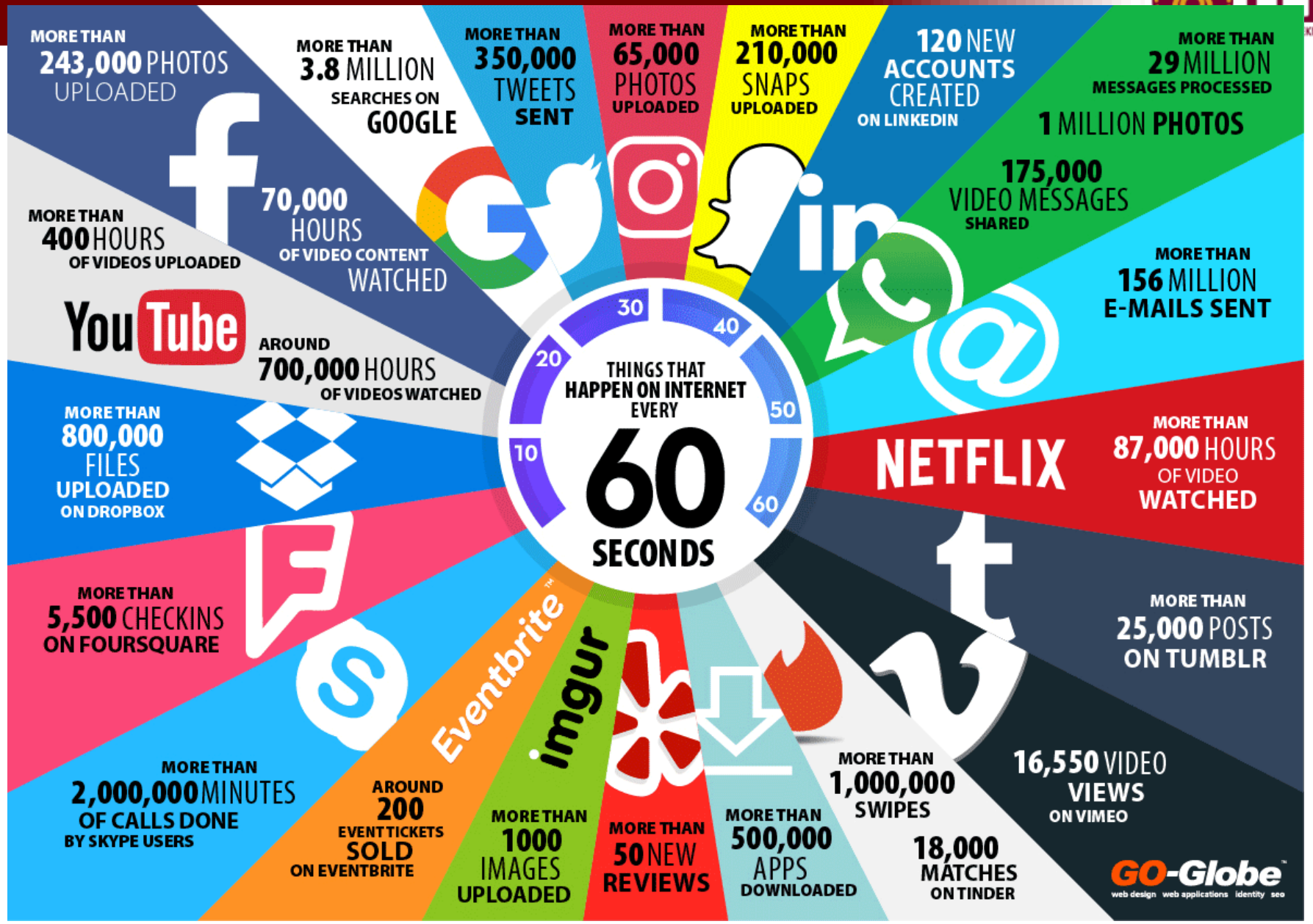
The 3 V's of Big Data



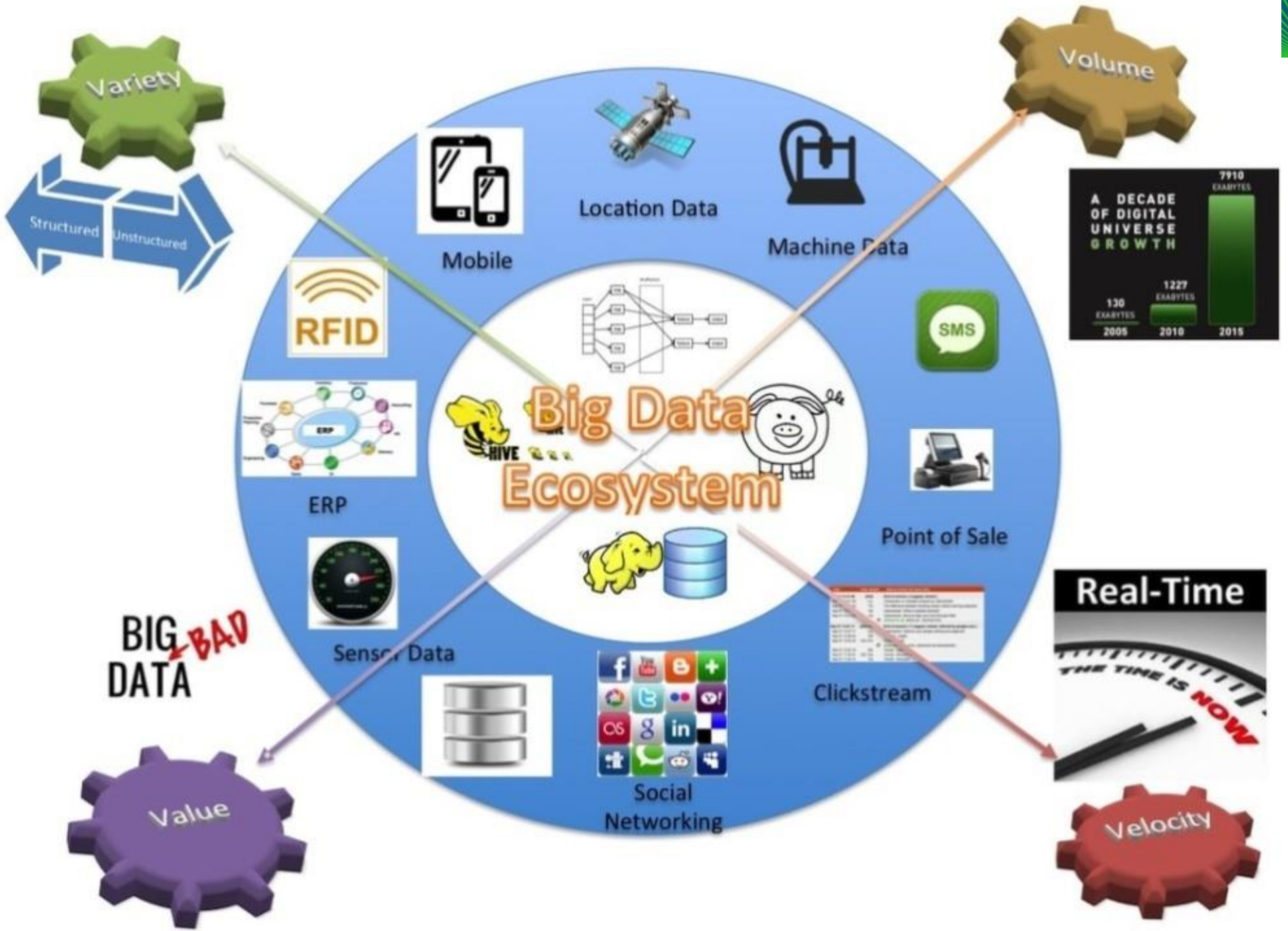
Big Data Market Projection | 2014 – 2026 ($B)



Data Source: Wikibon Big Data Project, 2016


MORE THAN 243,000 PHOTOS UPLOADED
MORE THAN 3.8 MILLION SEARCHES ON GOOGLE
MORE THAN 350,000 TWEETS SENT
MORE THAN 65,000 PHOTOS UPLOADED
MORE THAN 210,000 SNAPS UPLOADED
120 NEW ACCOUNTS CREATED ON LINKEDIN
MORE THAN 29 MILLION MESSAGES PROCESSED
1 MILLION PHOTOS
175,000 VIDEO MESSAGES SHARED
MORE THAN 156 MILLION E-MAILS SENT
70,000 HOURS OF VIDEO CONTENT WATCHED
MORE THAN 400 HOURS OF VIDEOS UPLOADED
YouTube
AROUND 700,000 HOURS OF VIDEOS WATCHED
THINGS THAT HAPPEN ON INTERNET EVERY 60 SECONDS
MORE THAN 800,000 FILES UPLOADED ON DROPBOX
NETFLIX
MORE THAN 87,000 HOURS OF VIDEO WATCHED
MORE THAN 5,500 CHECKINS ON FOURSQUARE
MORE THAN 25,000 POSTS ON TUMBLR
MORE THAN 2,000,000 MINUTES OF CALLS DONE BY SKYPE USERS
AROUND 200 EVENT TICKETS SOLD ON EVENTBRITE
MORE THAN 1000 IMAGES UPLOADED
MORE THAN 50 NEW REVIEWS
MORE THAN 500,000 APPS DOWNLOADED
MORE THAN 1,000,000 SWIPES
18,000 MATCHES ON TINDER
16,550 VIDEO VIEWS ON VIMEO
GO-Globe
web design web applications identity seo

BIG DATA


Variety
Structured
Unstructured
Volume
A DECADE OF DIGITAL UNIVERSE GROWTH
130 EXABYTES 2005
1227 EXABYTES 2010
7910 EXABYTES 2015
Location Data
Mobile
Machine Data
RFID
SMS
Big Data Ecosystem
ERP
Point of Sale
Sensor Data
Clickstream
BIG DATA BAD
Real-Time
THE TIME IS NOW
Social Networking
Value
Velocity

CHALLENGES IN BIG DATA ANALYTICS

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UTM BIG DATA CENTRE
TOGETHER WE MAKE MOTIONS...

What types of decisions do we need to make at strategic, tactical, operational management levels?





What types of INSIGHT can be used to uncover hidden patterns, unknown correlations and other real- time insights - Descriptive Diagnostic, Predictive, Prescriptive?

What knowledge can support the work of our knowledge workers?



How can we improve our business?



How can we clean, extract and integrate those data?

What types of real-time visualization can give insights to our business?



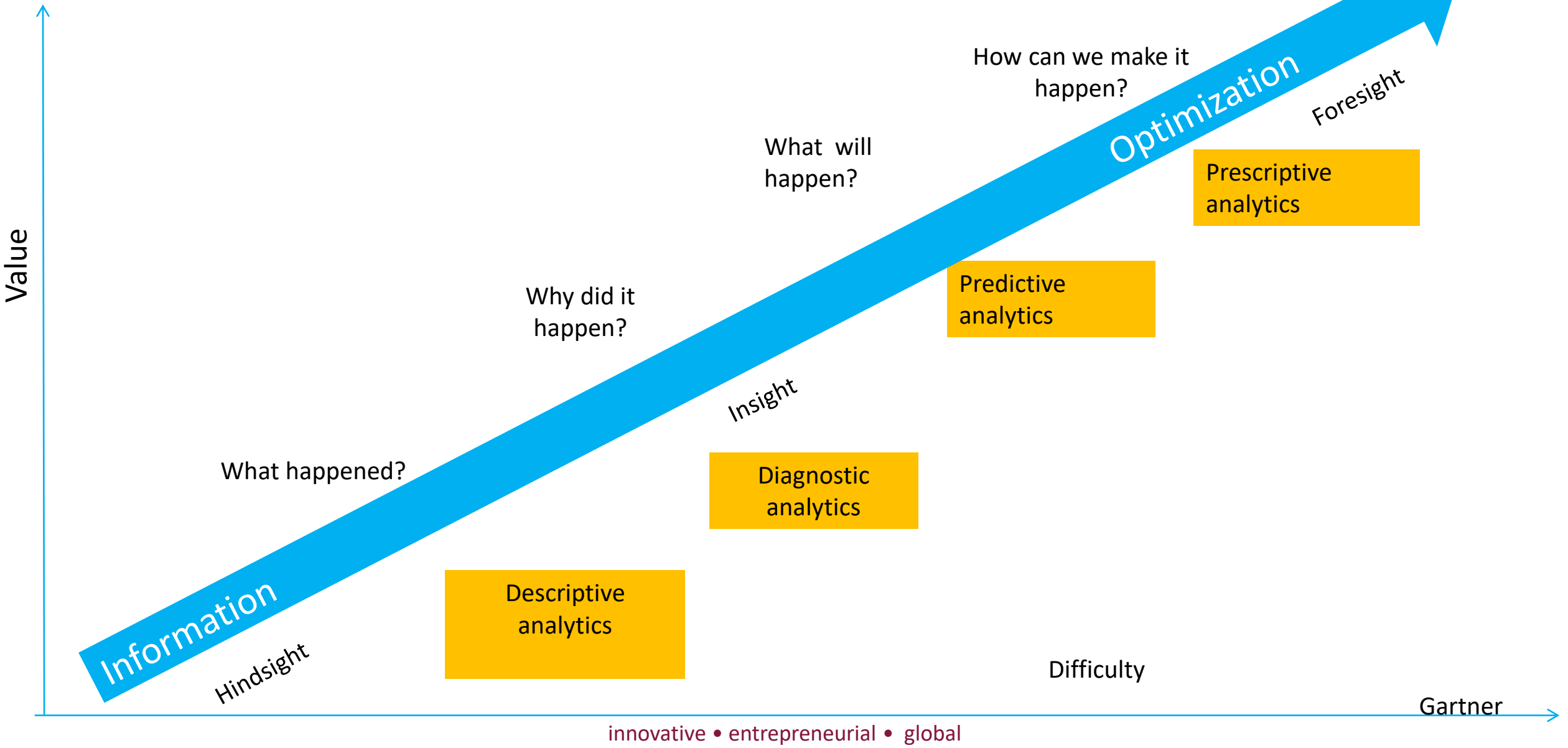




What sources of data can be used to produce those analytics ?

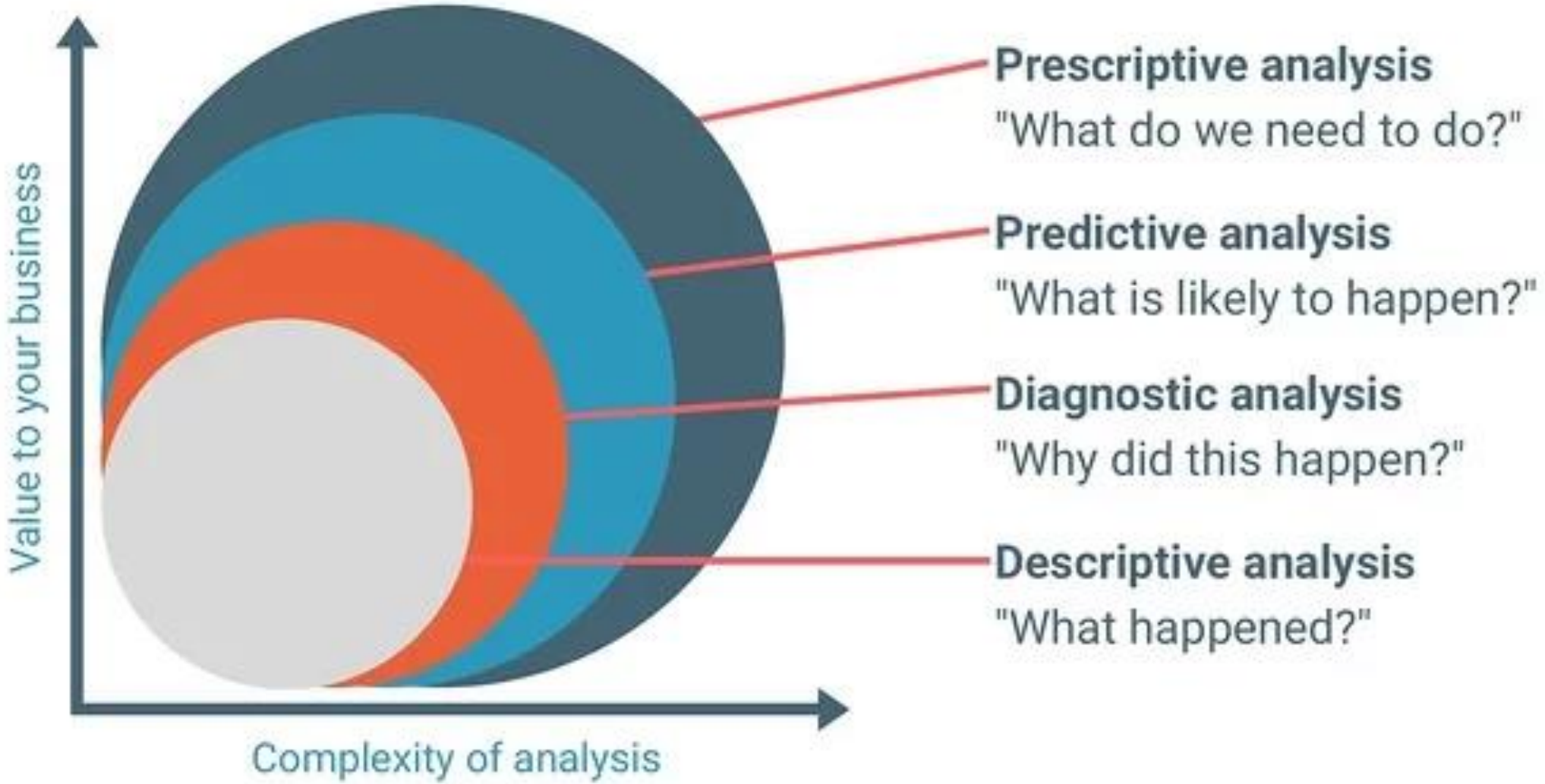
Can open data or data scraped from social media or the web be used?

Analytics Capabilities



4 Types of Big Data Analytics

G2 CROWD



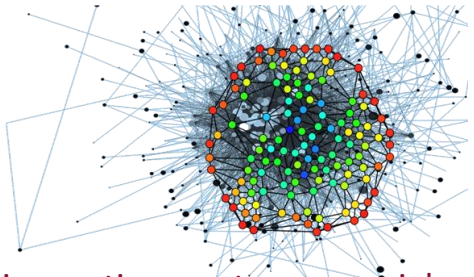
Big Data Analytics



Structured or Unstructured



Statistical Methods
Machine Learning
Artificial Intelligence…



innovative • entrepreneurial • global

Identify Patterns
Predict & Forecast
Optimization
Decision Making



DATA ANALYTICS APPLICATIONS

EACH SOLUTION IS UNIQUE

Customer Insight



Smarter Healthcare



Science & Research



m/c performance



Business Insight



Traffic Control



Retail Solutions



Finance



Personal Insight



Sports Performance



Homeland Security

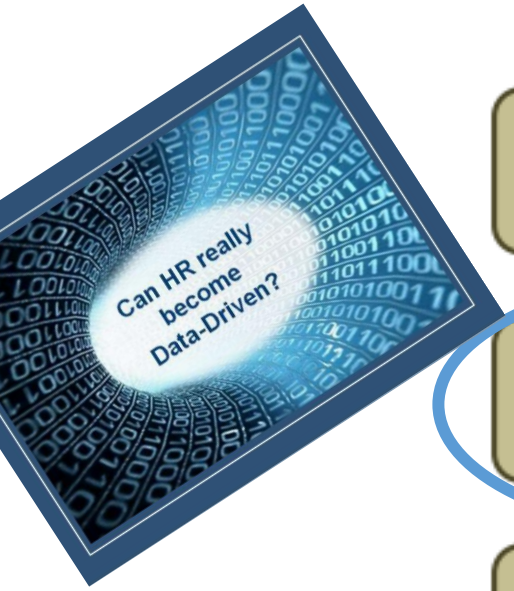


Risk Management



- Example
- **Predictive Analytics in HR @ HR Analytics**

Analytics Domains



- Retail Sales Analytics
- Financial Services Analytics
- Risk & Credit Analytics
- Talent / HR Analytics
- Marketing Analytics
- Consumer Behavior & Cohort Analytics
- Collections Analytics
- Fraud Analytics
- Pricing Analytics
- Telecom Analytics
- Supply Chain Analytics
- Web Analytics


Predictive
Analytics

UTM

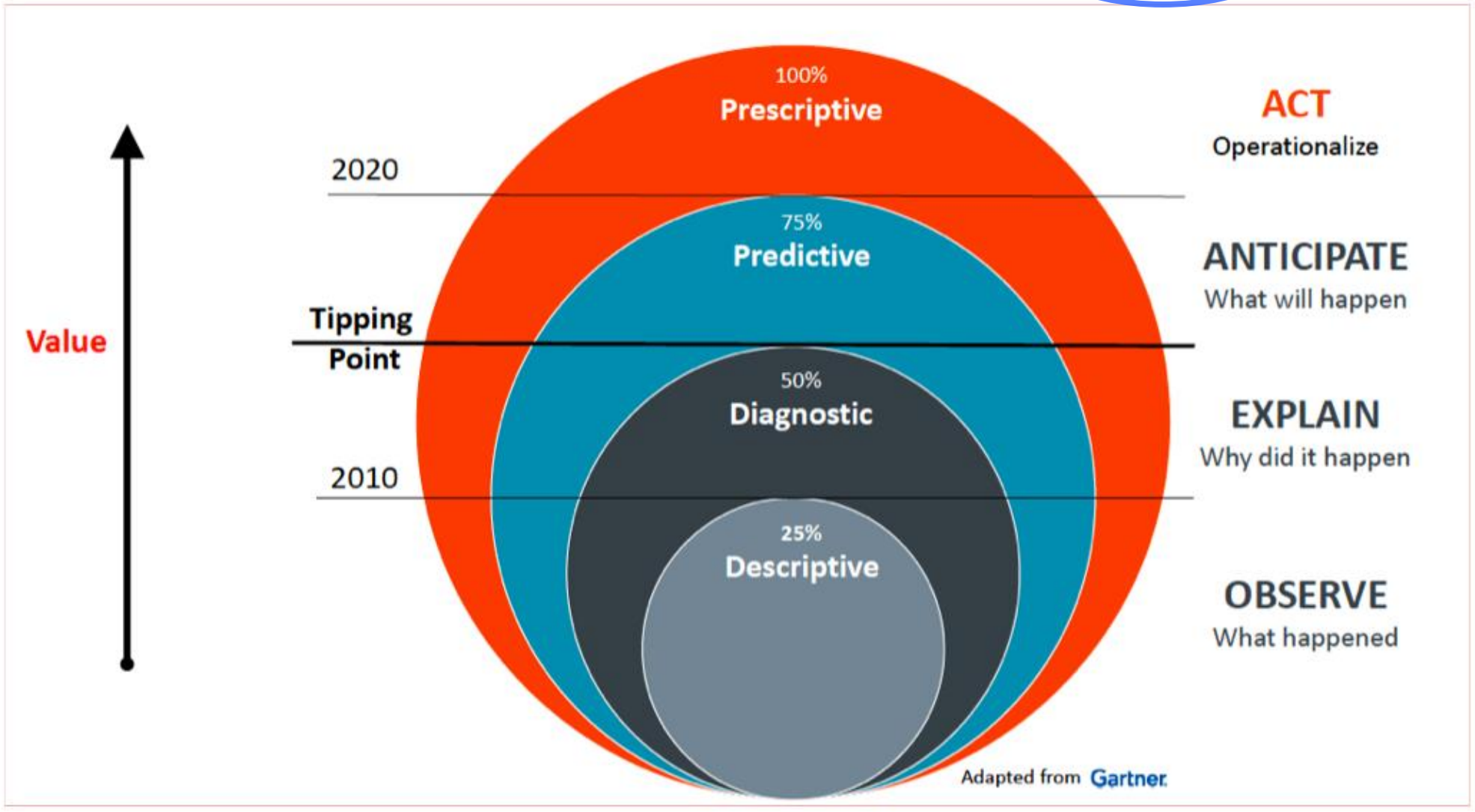
UTM BIG DATA CENTRE
TOGETHER WE MAKE MOTIONS...

Improving HR with Predictive Analytics

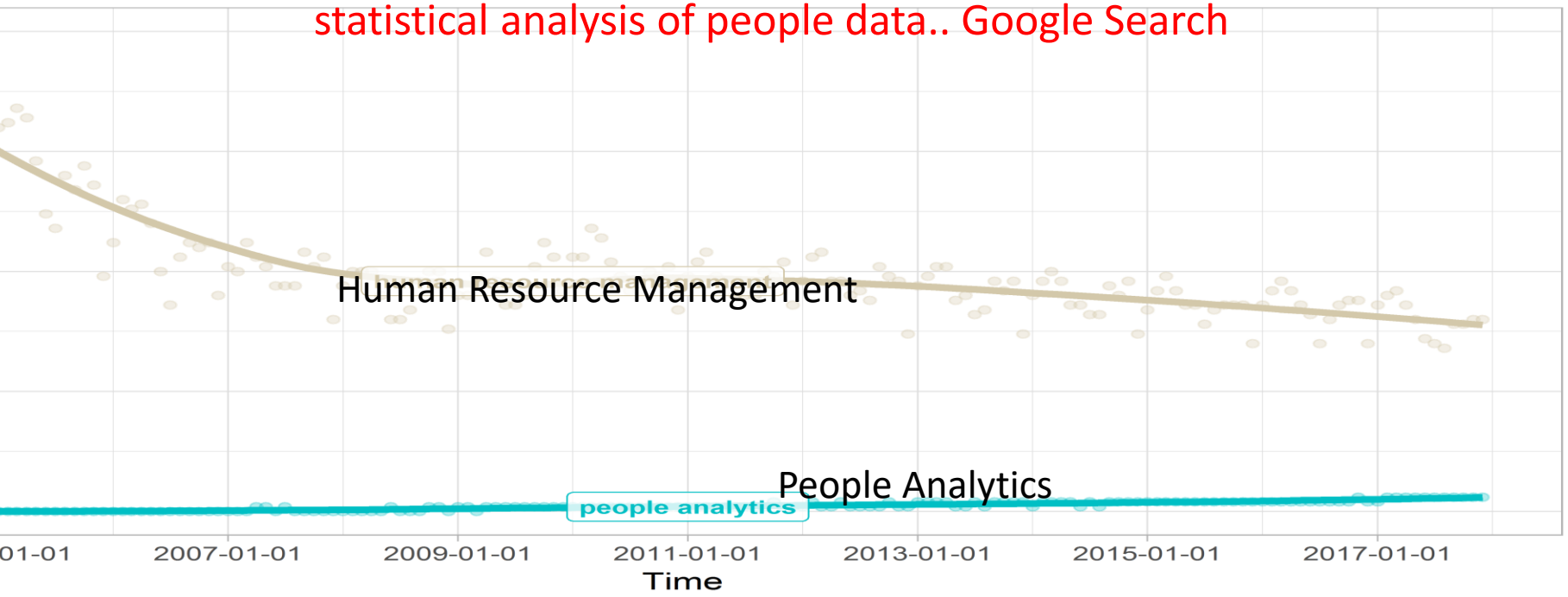


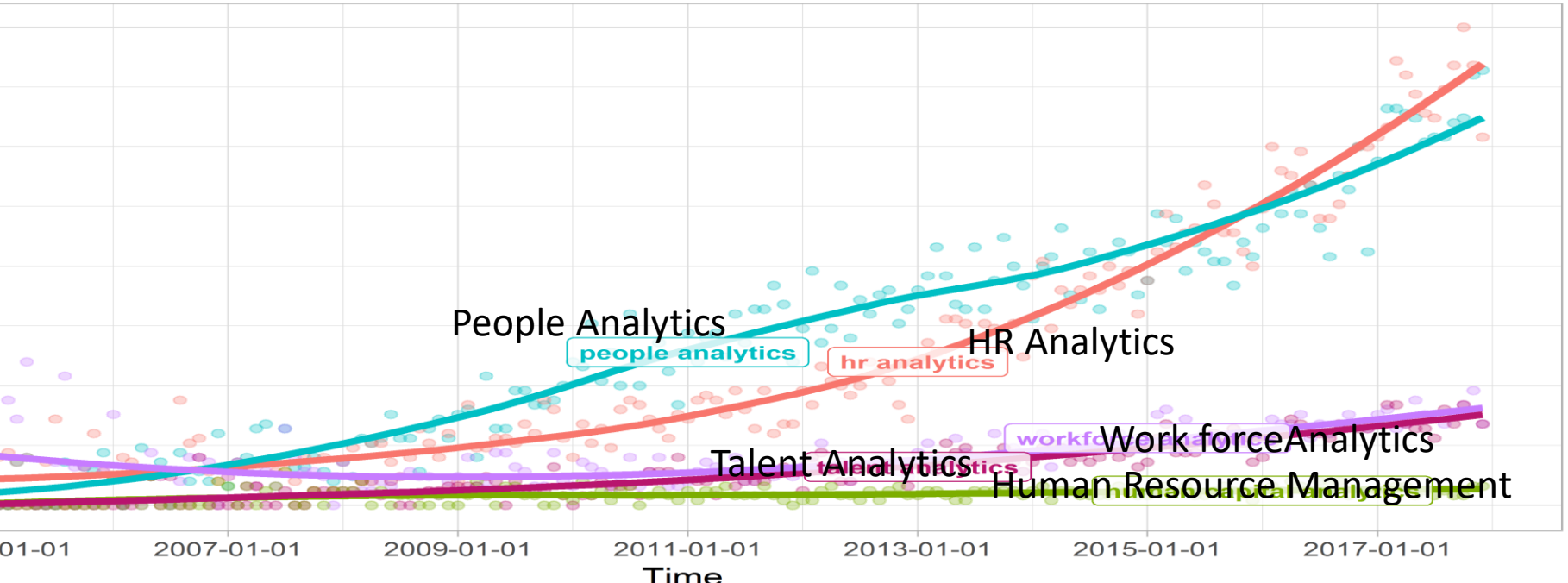
innovative • entrepreneurial • global

Predictive Analytics Transforms Insights into Action



statistical analysis of people data.. Google Search







Predictive Analytics in HR

Talent analytics has a slightly more exclusive connotation: it focuses on talent and was often used in academia

Prediction Scenarios in HR



Who would be the best fit out of 3 candidates for the job?



Of the two contenders, who would be better performer?



Which employees are likely to quit in the next 6 months?



Individual Level





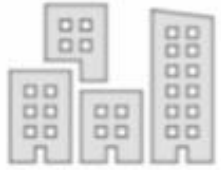
Selection and engagement process can be customized by using these information



Proactively track attrition to make sure that the top talent does not leave the organization

Prediction Scenarios in HR

1. Based on business growth numbers, what should be our headcount for next year?
2. Which locations should we increase headcount in the coming year to minimize turnover?
3. What training programs should we invest in, to get maximum customer satisfaction?
4. What would be the impact of new policies on employee engagement?
5. What will be the impact of the salary increase on retention?



Organization Level

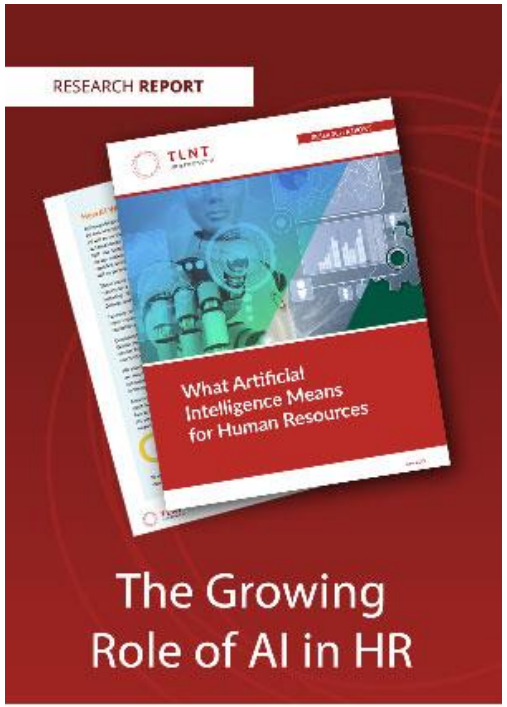


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What is Predictive Analytics

Simply: Using facts to help make decisions

- Identify an outcome you want to predict
- Look at data around the outcome (inputs)
- Use techniques to find and validate patterns in the data that predict outcome
- Deploy the “model” into daily decision making
- Model learns based on how it does in real life



Predictive Analytics



Predictions can be made about future events by analyzing current and historical data



Predictive models exploit patterns found in historical data to identify risks and opportunities



They capture relationships among many factors to allow risk assessment and guide decision-making



Leverages a variety of statistical techniques like predictive modeling, machine learning and data mining



Multiple statistical techniques like regression, logistic regression, decision trees etc., are used


Core HR
Succession Planning
Absence
HR Analytics
Learning
Payroll
Performance
Recruitment
Attrition


ANALYTICS

innovat


EMPLOYEE ENGAGEMENT


HR DECISION ARE NO LONGER BASED ON GUT FEELING
Operational HR
1900s
Strategic HR
2000s
Data-driven HR
Now
AIHR
BLOG & ACADEMY

Predictive Analytics: Forward-Looking Innovation



- *Data & algorithms don't care* about gender, skin color, education or other personal details.
- Care only about the ability to predict an outcome

Data-Driven-Decision



ABC

Age

Gender

Education

Number of companies worked for in the past

Department

Job level

Stock option ownership

Recent performance rating

Total years of experience

Years in current organization

Years since last promotion

Years with same manager

Last salary hike %

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DATA-DRIVEN DECISION MAKING IN HR

- demographics
- pay
- engagement
- performance
- ATS

Combine & Clean

Data is exported from existing systems and combined in one dataset...

Analyses & Data Mining



...which is used to analyze data. By analyzing patterns and causal relations one can build predictive models.

AIHR
BLOG & ACADEMY

Can We Predict Great Leadership?

predictive analytics to analyze the correlation between leadership behavior and business outcomes

- Identify the business goals and corresponding performance objectives. Pick two or three key objectives.
- Define the desired behaviors of leaders. Be specific about the behaviors your company needs in its leaders. Ensure these behaviors align with the unique culture of your organization.
- Gather data on the activities that will isolate and develop the factors that demonstrate a strong correlation to leadership success.
- Determine the data and measurements you will need from the outset. Find a skilled data analyst if you do not have that capability internally.
- Focus your leadership development so it has the greatest impact. Once you have correlated the data with your business metrics and isolated the effect of learning on your desired leadership behaviors, use the results to refine your approach.



How Predictive Analytics Can Make Talent Development More Effective

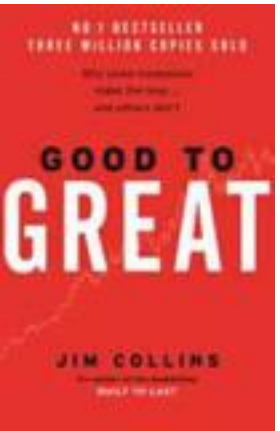
- In an ideal business state, you are able to dedicate the appropriate time and resources to recruiting, hiring, and developing key individual contributors and current and future leaders, and such efforts are carried out systematically.
- In an ideal business state, you're ahead of the game and make the best possible talent decisions instead of scrambling to keep up and making unwanted concessions and compromises.
- The actual business state, unfortunately, is a lot messier.
- Predictive Analytics -an integral part of talent development strategy, adding consistency to the process, limiting human bias, and identifying organizational talent gaps



Talent Analytics



- practical application of big data techniques to human resources
- taking into account the data of past, current, and potential employees to produce insight. With this insight, businesses can be more easily identify top employee qualities and use this valuable information to hire precisely the right candidates.
- The ability to attract talent — the right people, not just anyone – is now one of the biggest differentiating factors in the business world today. find people who “fit” – fit within their strategies, their cultures, and their teams, as well as the open jobs themselves.
- With the right data in hand, recruiting the best candidates is not only faster, but also more cost-effective and objective.
- Analytics technology is bringing about a transformation that is gradually changing HR’s function entirely, making it more integral to business goals and strategies than ever before.



insights into whom to hire

widening the pool of candidates

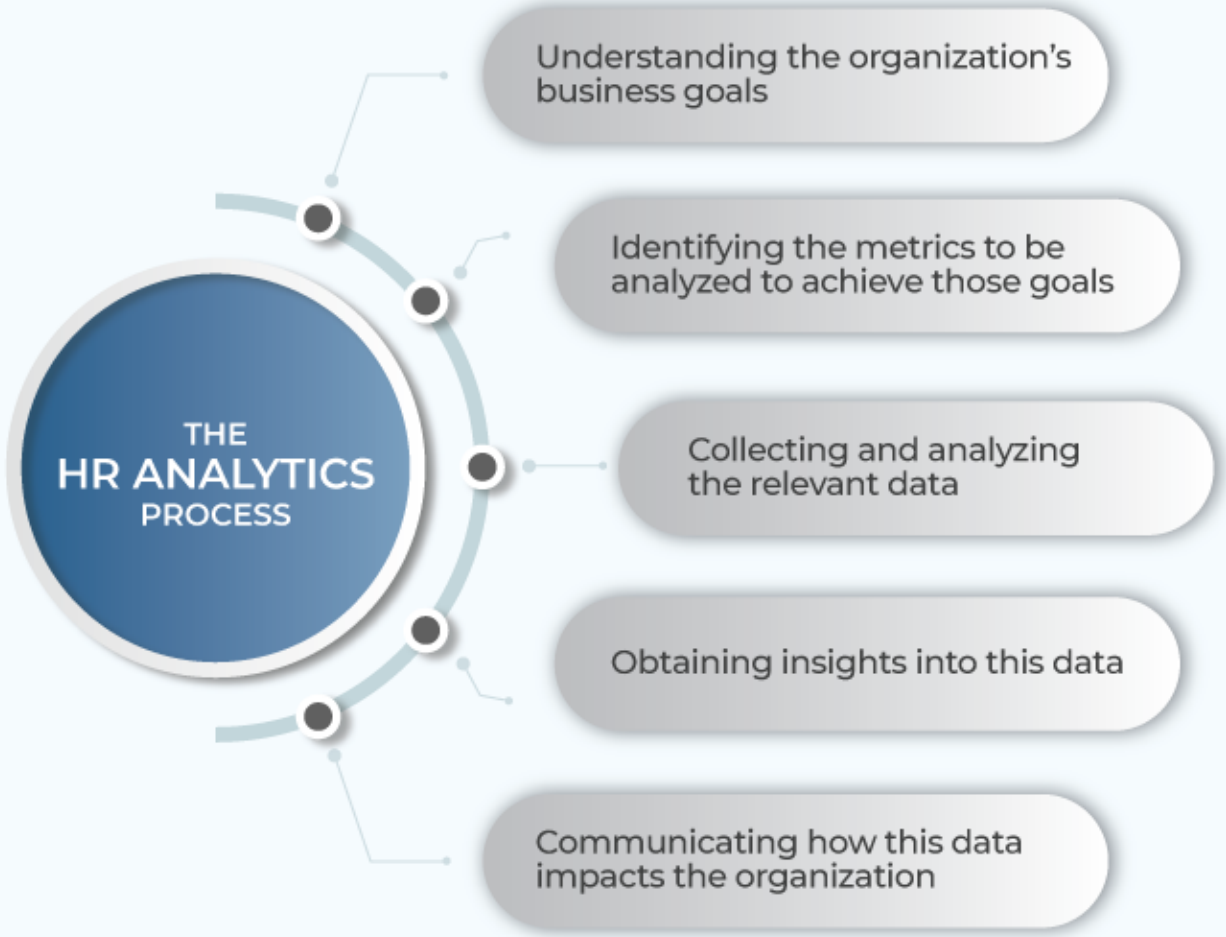
conduct more effective succession planning

stay on top of certifications and training

help companies better analyze turnover

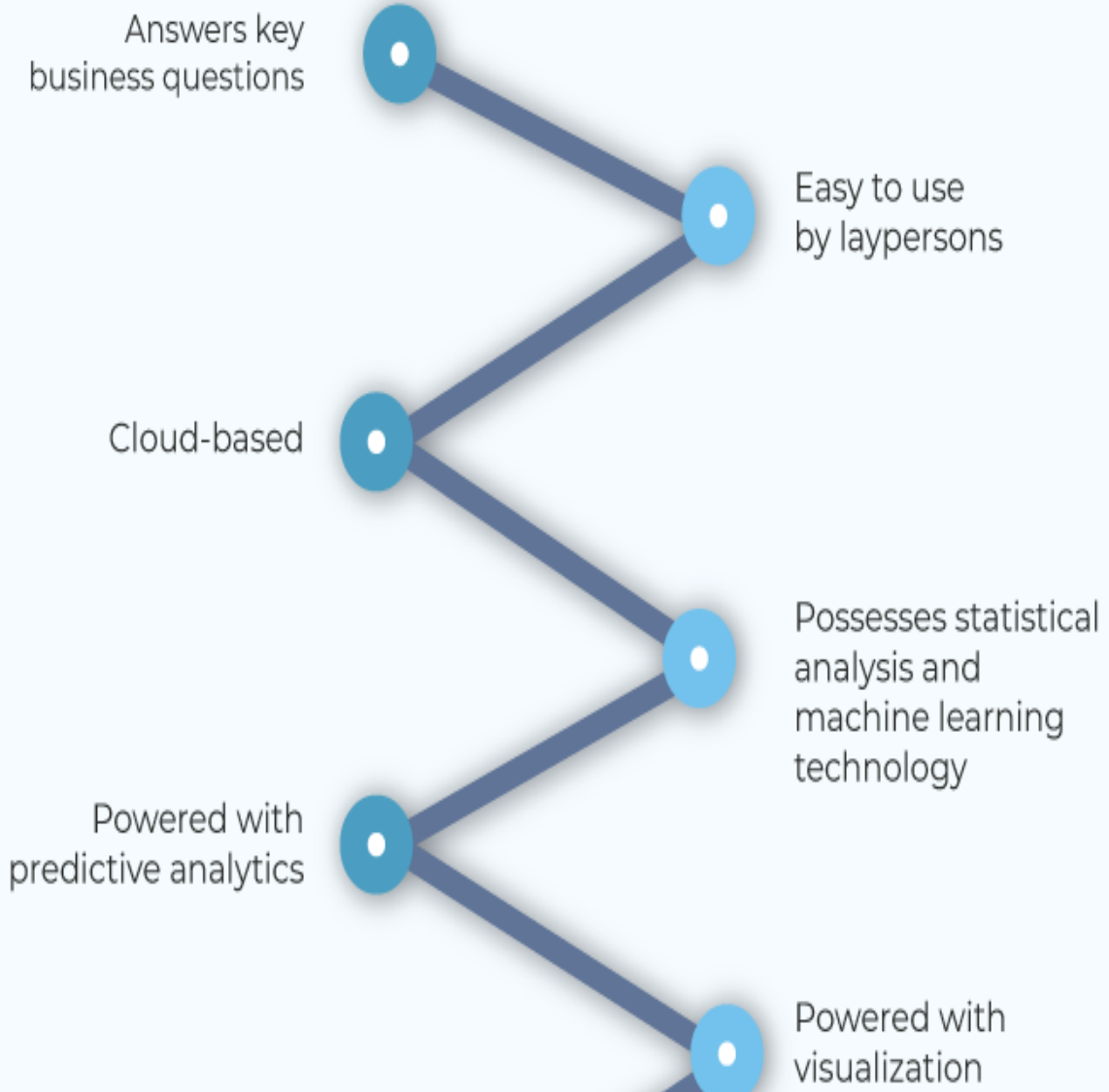
identify top performers ensuring they are effectively engaged

THE HR ANALYTICS PROCESS

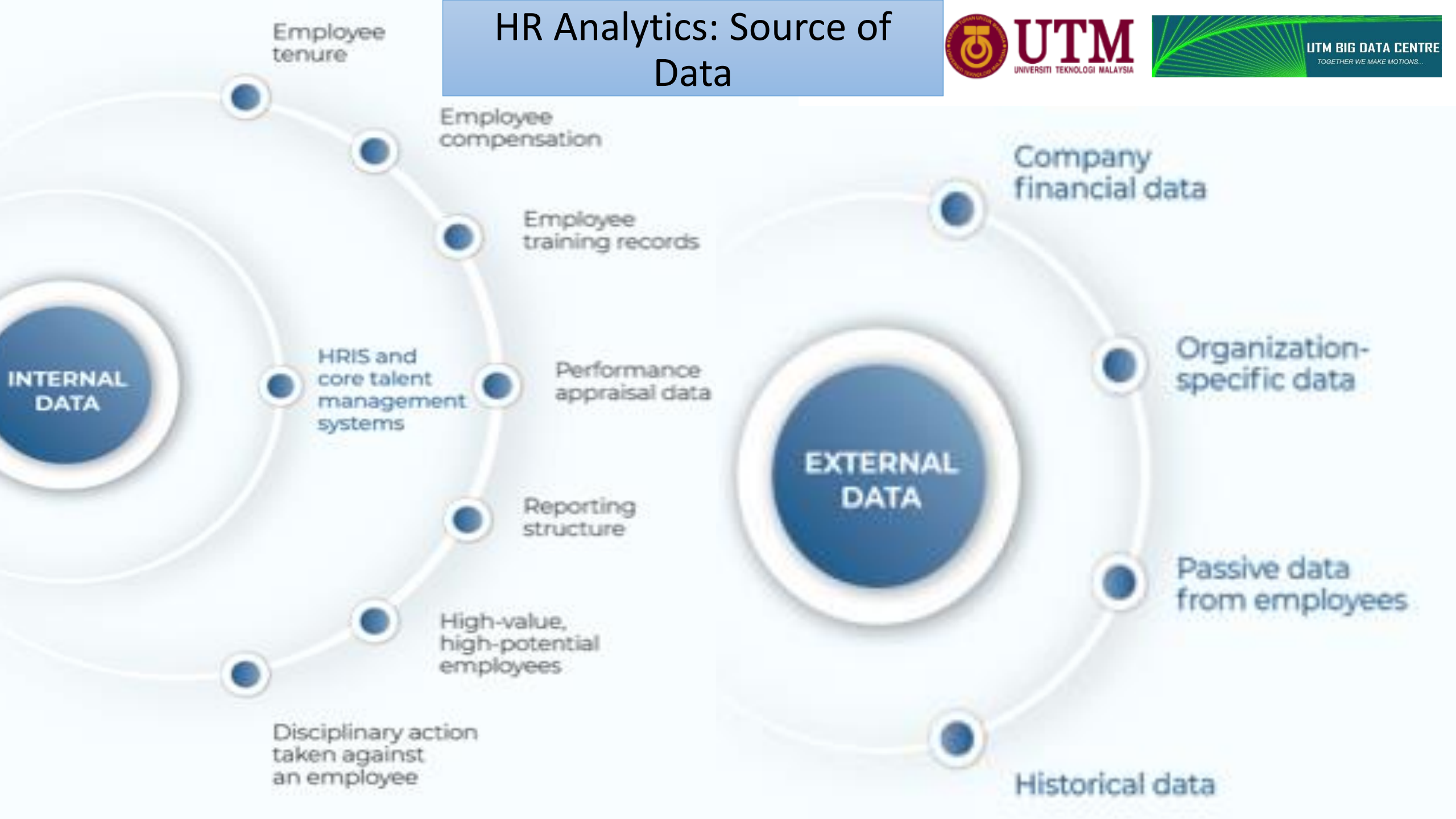


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FEATURES OF A GOOD HR ANALYTICS SOLUTION



HR Analytics: Source of Data



Steps in HR Analytics

Clip slide







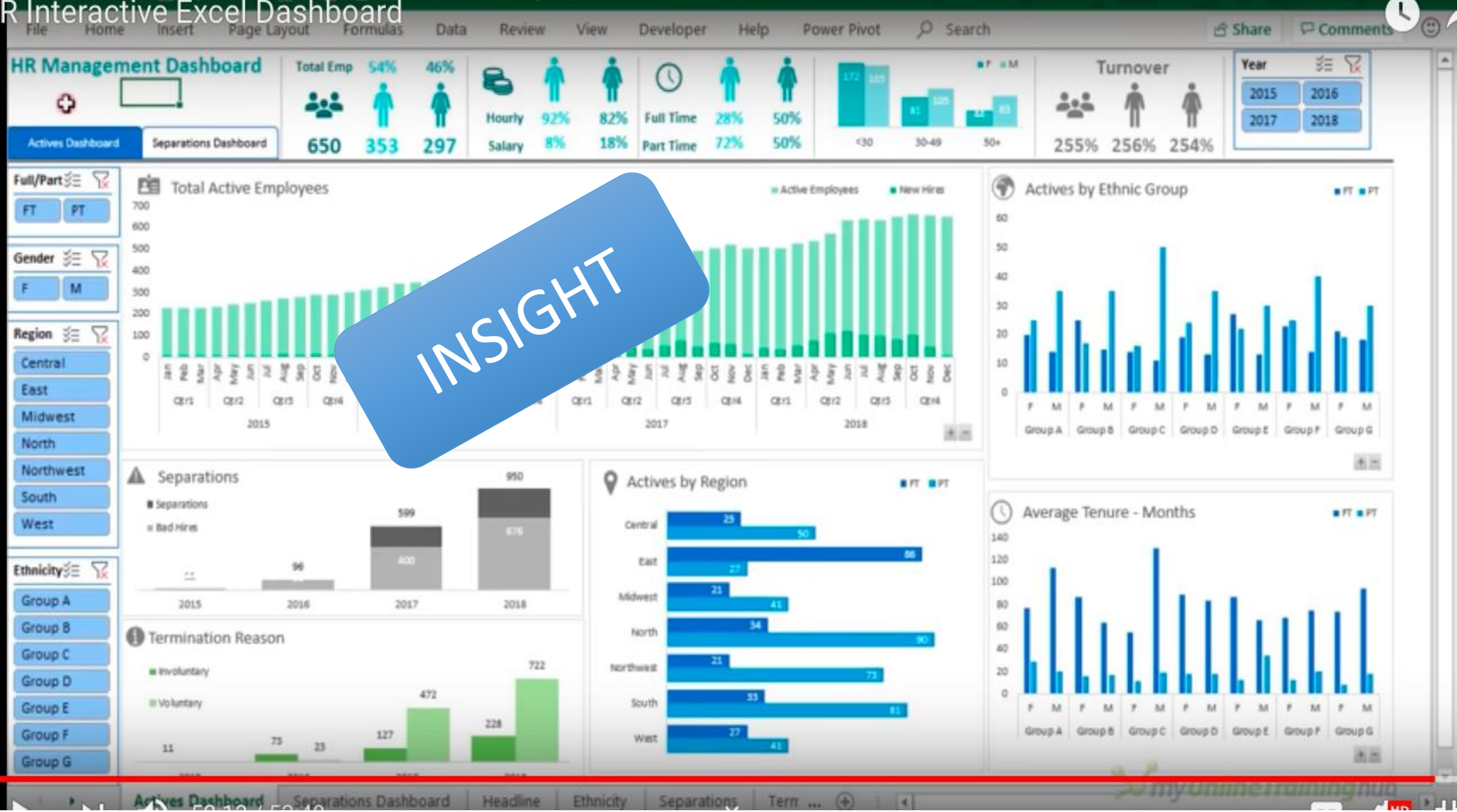
Knowing that you need **strategic thinkers** is one thing; knowing how to hire them is quite another.

G1 | Performance Rating

	A	B	C	D	E	F	G
1	Employee Number	Employee Status	Last status change	Date of Joining	Role	CTC (INR)	Performance Rating
2	A001	Active	Promotion	8/25/2010	Lead	800000	Meets Expectations
3	A002	Active	Promotion	1/18/2010	Manager	1200000	Meets Expectations
4	A003	Active	Appraisal	5/16/2011	Engineer	550000	Meets Expectations
5	A004	Active	Appraisal	6/15/2010	Lead	850000	Exceeds Expectations
6	A005	Active	New join	5/10/2013	Engineer	555000	Outstanding
7	A006	Active	New join	5/11/2013	Engineer	560000	Exceeds Expectations
8	A007	Active	New join	5/12/2013	Engineer	565000	Meets Expectations
9	A008	Active	Appraisal	2/15/2010	Engineer	570000	Meets Expectations
10	A009	Active	New join	7/11/2013	Lead	1030000	Meets Expectations
11	A010	Active	Promotion	1/5/2	Director	2200000	Below Expectations
12	A011	Active	New join		Lead	1125000	Meets Expectations
13	A012	Active	Promotion		Lead	965000	Meets Expectations
14	A013	Active	Promotion		Manager	1800000	Meets Expectations
15	A014	Active	Promotion		Manager	1400000	Meets Expectations
16	A015	Resigned	Ex		Engineer	575000	Exceeds Expectations
17	A016	Resigned	E		Engineer	580000	Meets Expectations
18	A017	Active	Appra	0	Engineer	585000	Outstanding
19	A018	Active	Appraisa	15/2010	Engineer	590000	Meets Expectations
20	A019	Active	Appraisal	6/2/2010	Engineer	595000	Meets Expectations
21	A020	Active	Appraisal	9/5/2011	Engineer	600000	Exceeds Expectations
22	A021	Active	Appraisal	9/24/2012	Engineer	605000	Meets Expectations
23	A022	Active	Appraisal	10/18/2010	Engineer	610000	Below Expectations
24	A023	Active	Appraisal	11/3/2010	Engineer	615000	Meets Expectations
25	A024	Active	Appraisal	3/22/2010	Lead	1150000	Exceeds Expectations
26	A025	Active	Appraisal	7/6/2011	Manager	1600000	Outstanding
27	A026	Active	Appraisal	6/30/2010	Engineer	620000	Exceeds Expectations
28	A027	Active	Appraisal	9/13/2010	Lead	950000	Meets Expectations
29	A028	Active	Appraisal	2/16/2011	Engineer	625000	Exceeds Expectations
30	A029	Active	Appraisal	6/14/2010	Engineer	630000	Below Expectations
31	A030	Active	Promotion	6/8/2011	Manager	1700000	Meets Expectations

HINDSIGHT

HR Interactive Excel Dashboard



About | **Dashboard** | HeadCount | Leavers | LastOpinion | Talent (current) | Talent & CEM | Recruitment | OnBoarder | Sickness & Absence

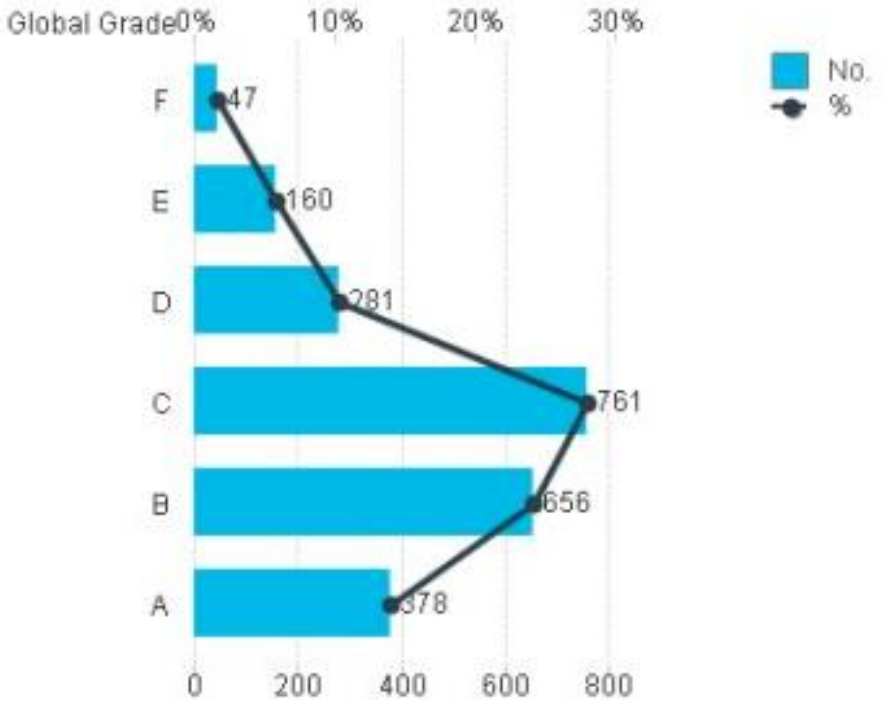
Dashboard

Current Year View 2015

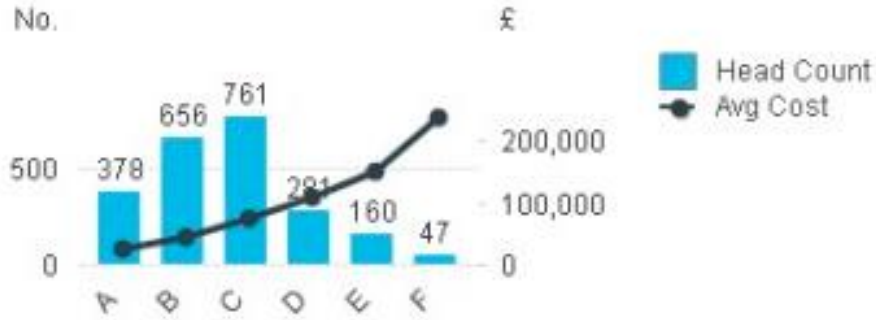
2014 | 2015 | 2016 | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec

Search

Headcount (active) by Grade



ARC by Grade



Currently viewed headcount 2283

Talent and CEM

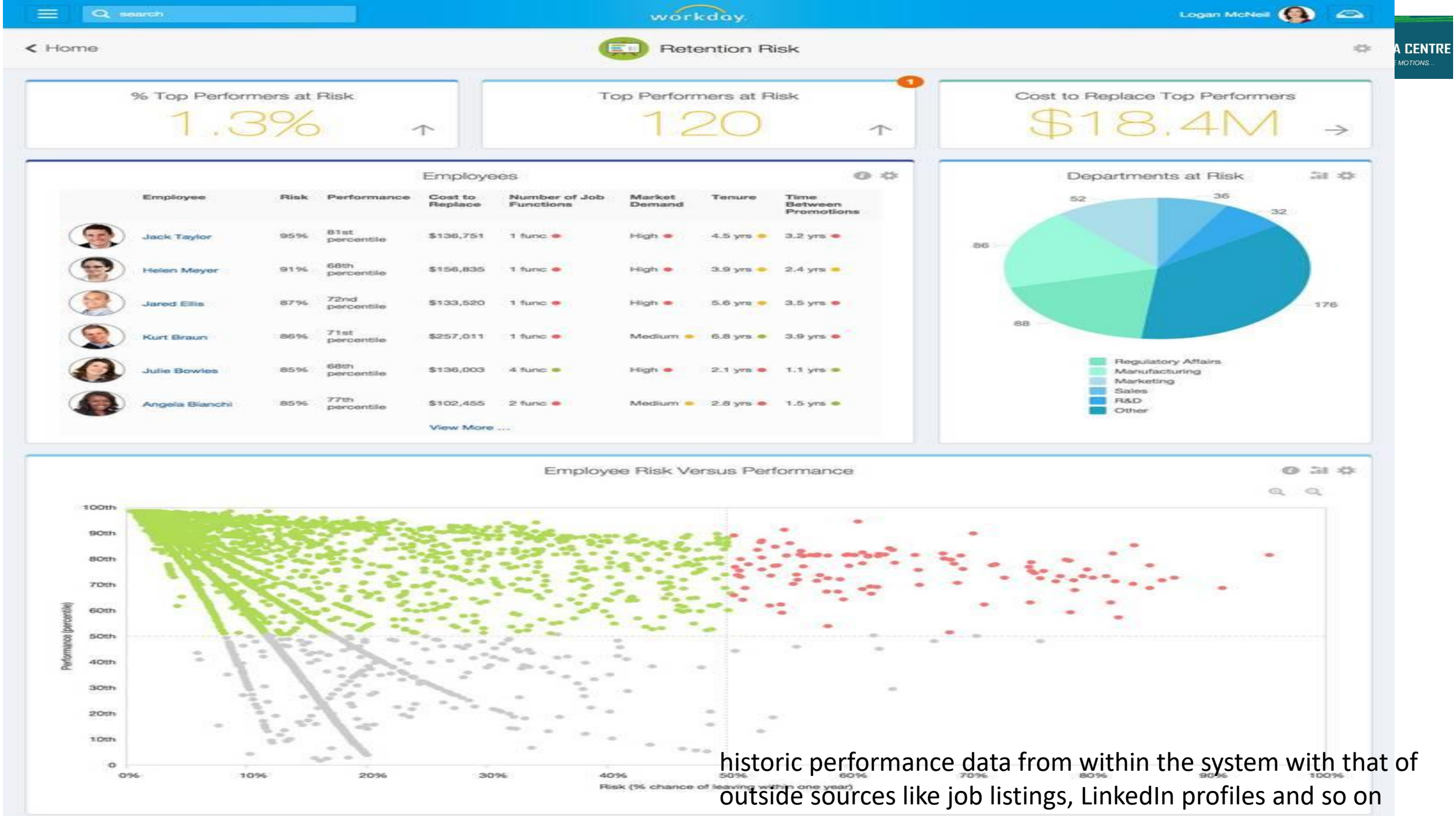
Top Talent	284	Global Off Track Year Start	352
Top Talent %	12.4%	Reduction	66
Top Performers	697	Global Off track YTD	286
Top Performers %	30.5%		

KPI Overview

Leavers	Joiners	Promotions	
424	447	371	21.1%
A Grade Hires	**College Hires**	**Av Cost of Hire**	**Cost of Hire %**
179 40.0%	158 35.3%	£6196.20	13.4%

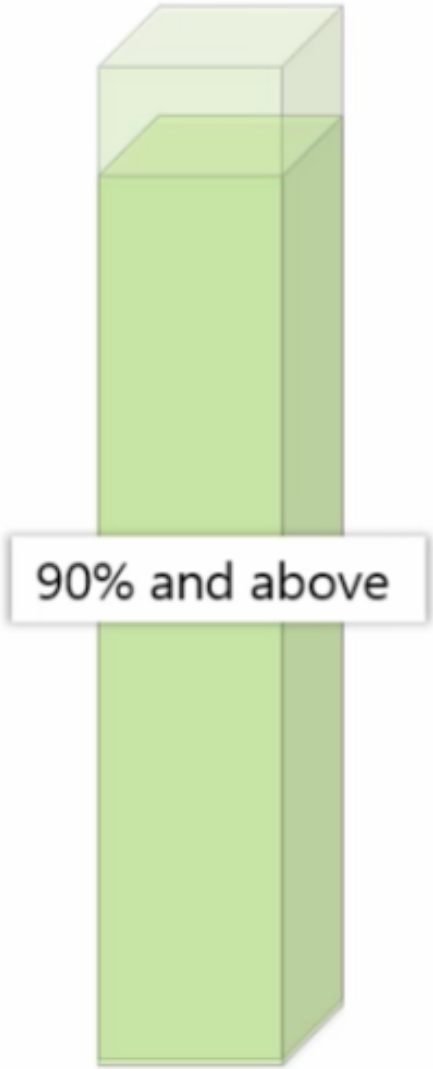
Leavers

Regretted	130	5.7%	Top Talent	41	1.8%
Non Regretted	294	12.9%	Top Performers	82	3.6%
Total Leavers	424	18.6%	Managed	181	7.9%



historic performance data from within the system with that of outside sources like job listings, LinkedIn profiles and so on

Assessing Accuracy of Model





A model which is able to predict with 2/3rd of the time with accuracy – everybody will be comfortable with it



Build the model



Train the model



Feed it back with incoming information

Assessing Accuracy of Model

Specificity and Sensitivity



FORESIGHT

Outcome for predicting attrition

	Actual	
	Stay	Leave
Predicted		
Stay	10	5
Leave	3	7

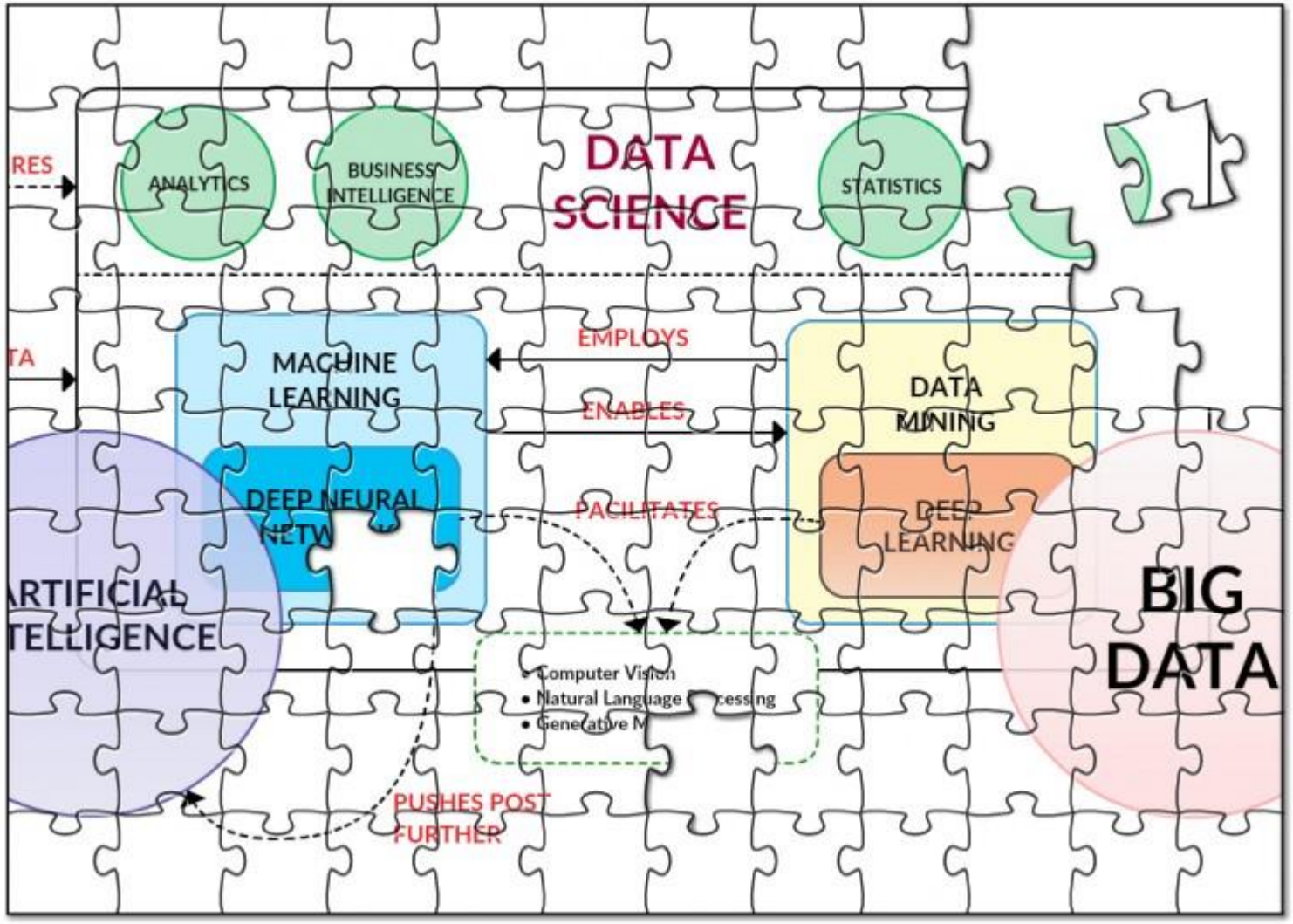
The model has clearly predicted:
- **10** as staying
- **7** as leaving

17 out of a total of **25** people have been accurately forecast

Accuracy = (10+7)/(10+7+3+5)= 17/25 = **68%**

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scout talent?



How AI is helping sports teams scout star players

Algorithms are transforming the way athletes are evaluated.



Talent Spotting?? The power of more data

- Spotting the next star athlete has always been as much art as science
- AI is transforming everything from business to healthcare now starting to muscle in on professional athletics too.
- Computer vision, machine learning and other forms of AI use algorithms to analyze player performance statistics, game videos, and data from various sensors to identify talent that coaches and scouts might otherwise miss.
- And since the algorithms **comb** through data far faster than humans can, they give teams in-depth information on more players than previously possible.
- Professional baseball, basketball and hockey are among the sports now using AI to supplement traditional coaching and scouting.
- Baseball scouts in particular have long used statistics to evaluate players.
- But now teams are starting to use AI to analyze an ever-expanding set of player data — including
  - radar gun data (throwing speed and spin),
  - video tracking (how players move around the field) and
  - swing speed and mechanics from sensor-studded bats.

HR.. Analytics

- The same algorithms that are helping us decide our next purchase on Amazon could help employers select the best of us from the candidates' shelf.
- Our education, experience, know-how, and network are becoming product features.
- For now, it is safe to assume that algorithms will be used just for screening purposes.
- Interviews, tests and personal interaction are still the primary recruiting tools, but analytics ensure that less promising candidates do not get an invitation, saving time and money.



THANKS



I ♥ UTM